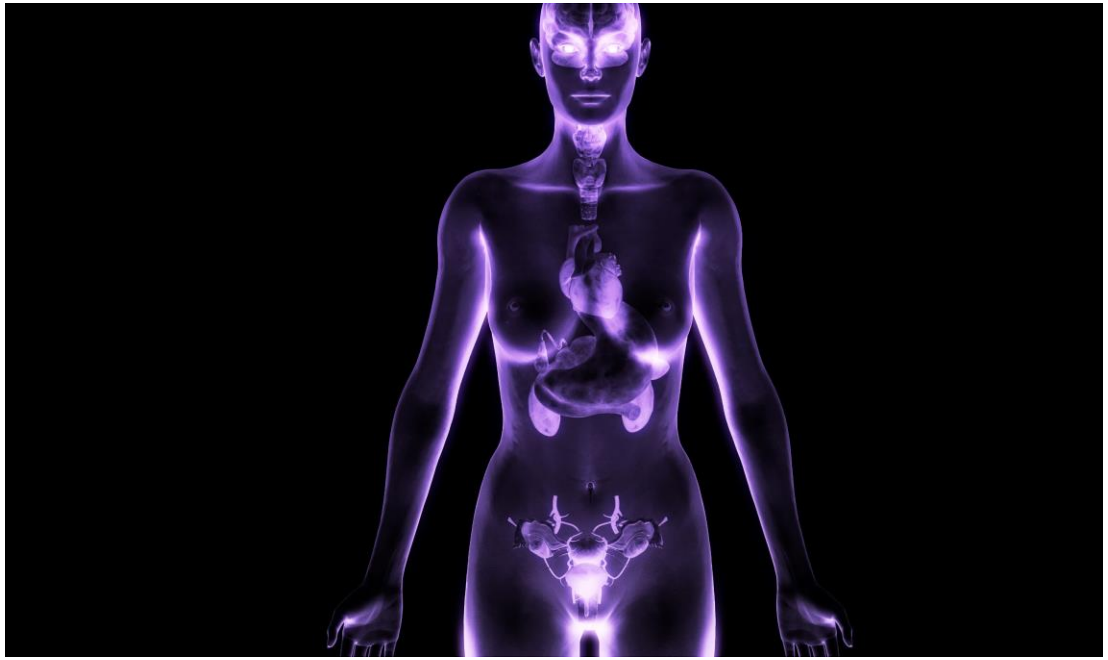
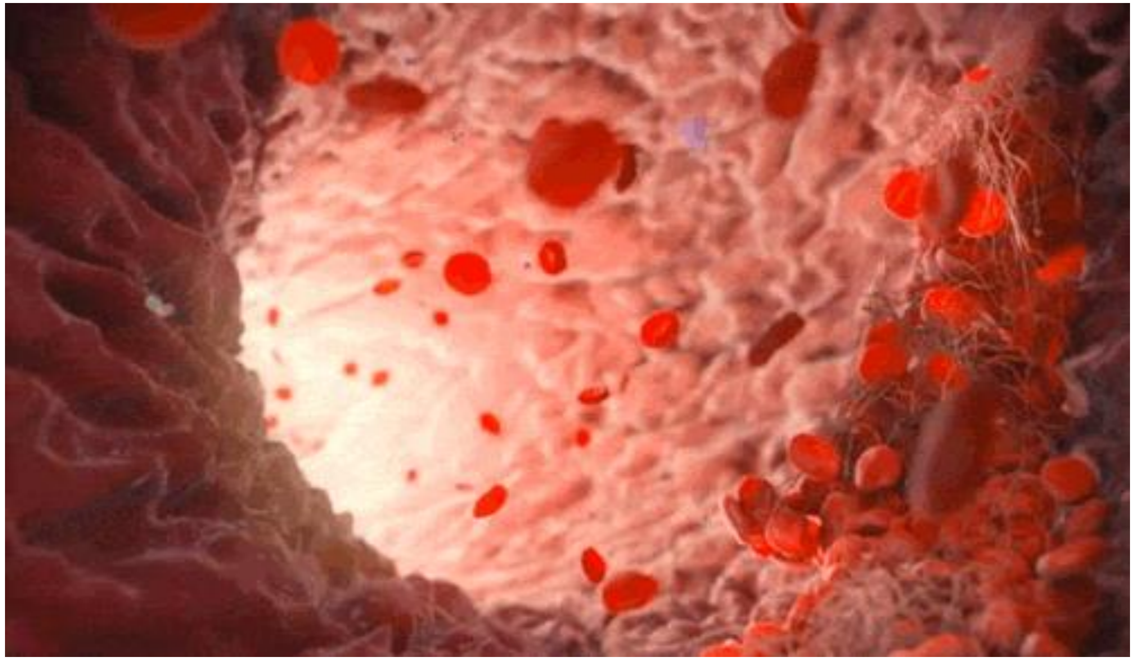
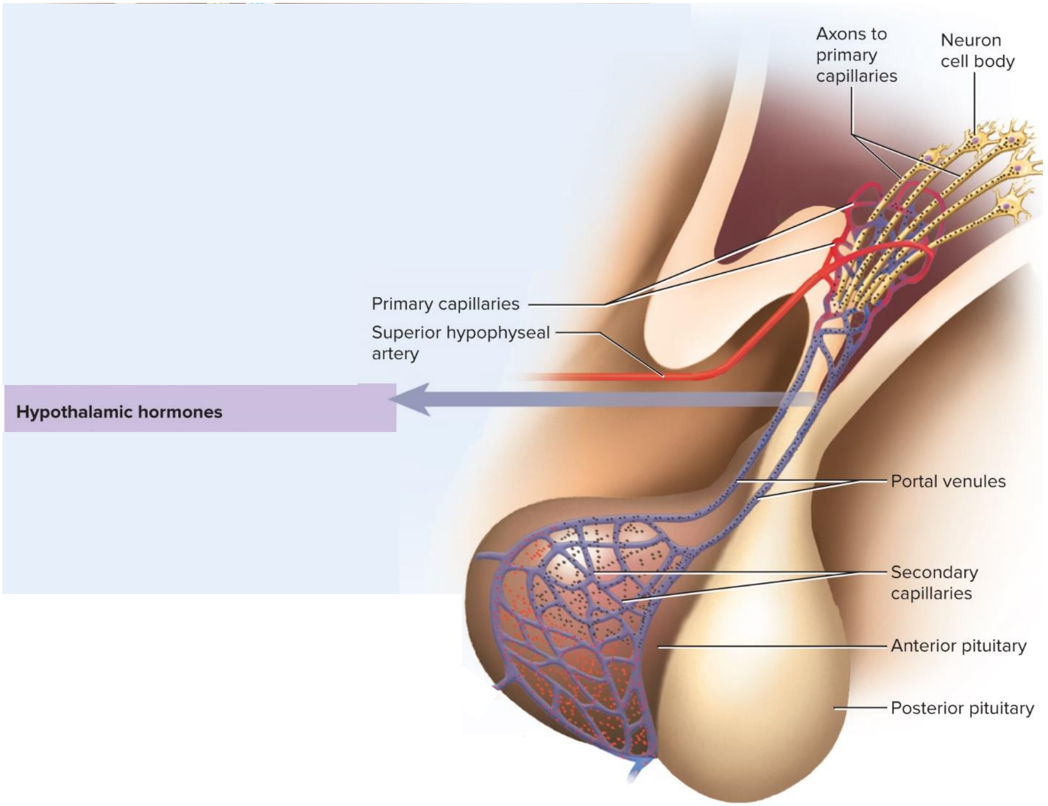
Exercise 13: Endocrine & Blood





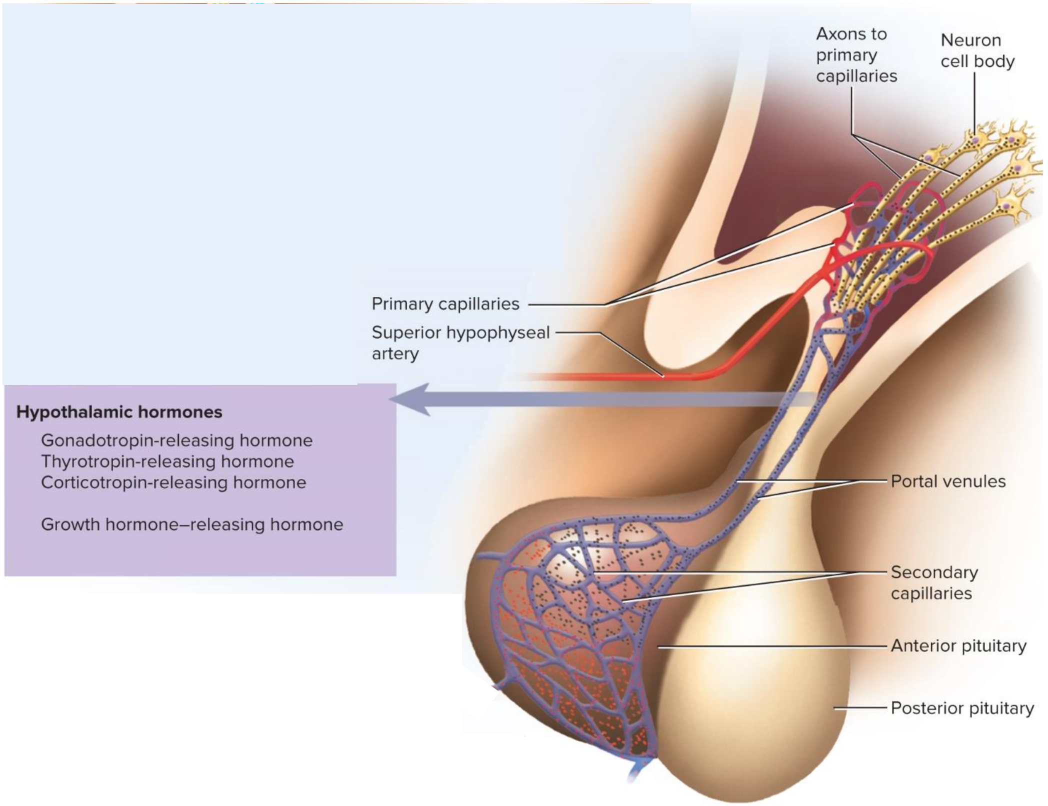
The Hypothalamus and Pituitary Gland



***hypothalamus** produces eight hormones

- six regulate the anterior pituitary and two are stored in the posterior pituitary

The Hypothalamus and Pituitary Gland



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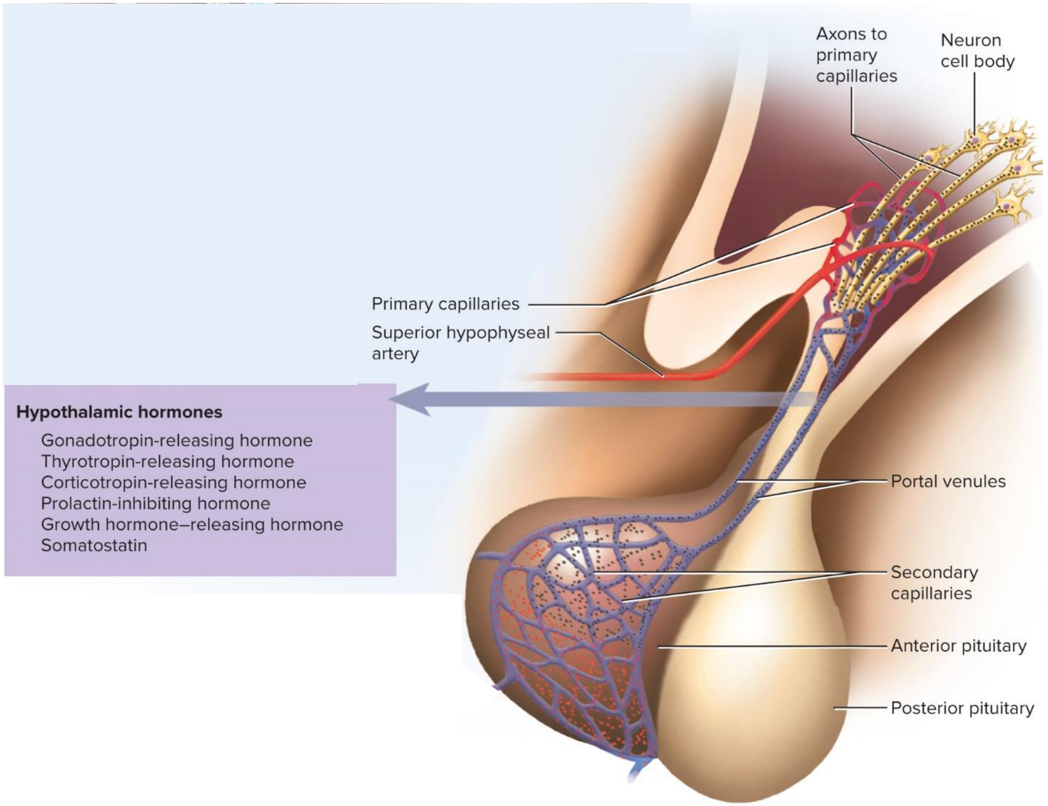

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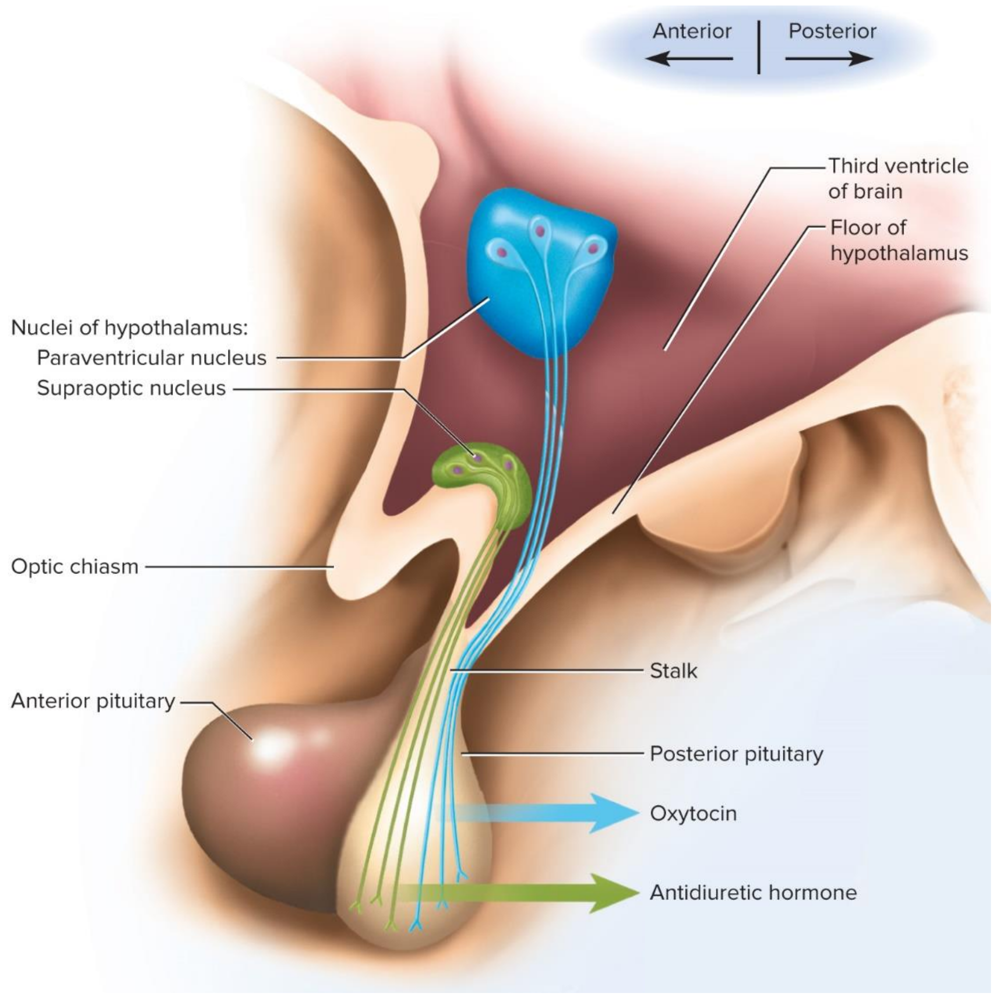

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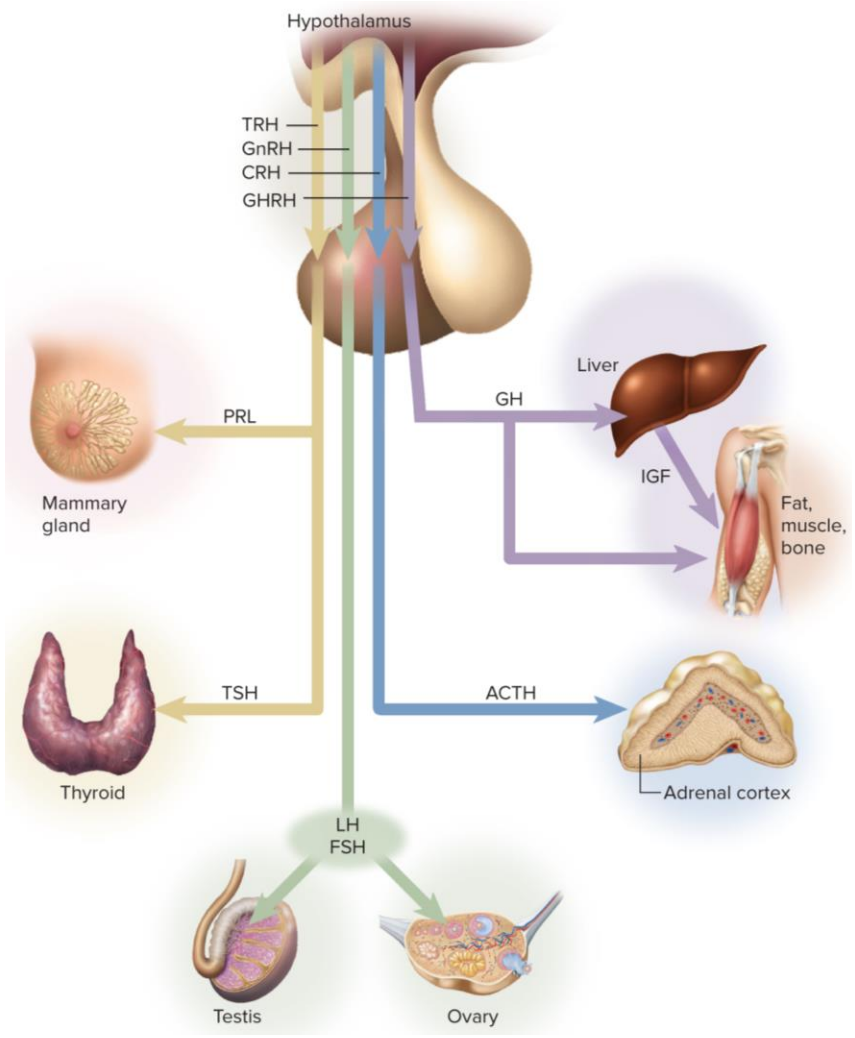
****oxytocin*** (***OT***) and ***antidiuretic hormone*** (***ADH***) are stored and released by the *posterior pituitary* neuronally

The Hypothalamus and Pituitary Gland



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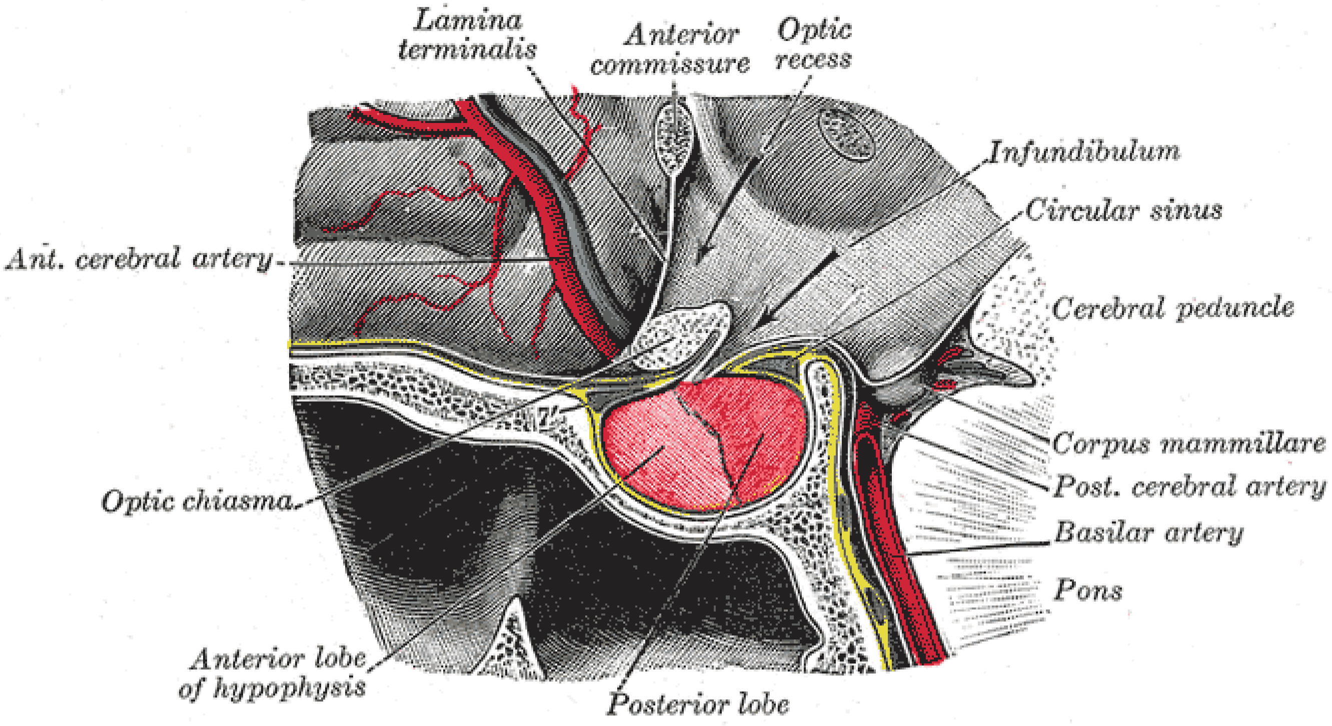
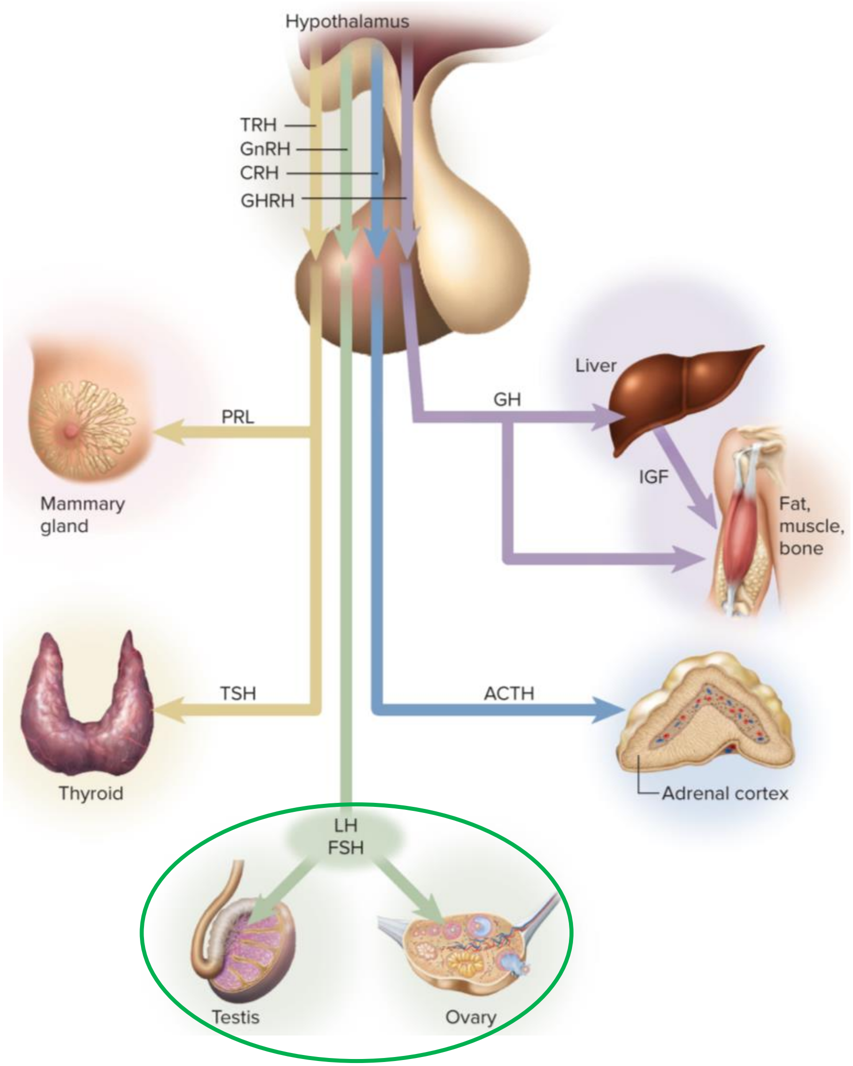


The Hypothalamus and Pituitary Gland



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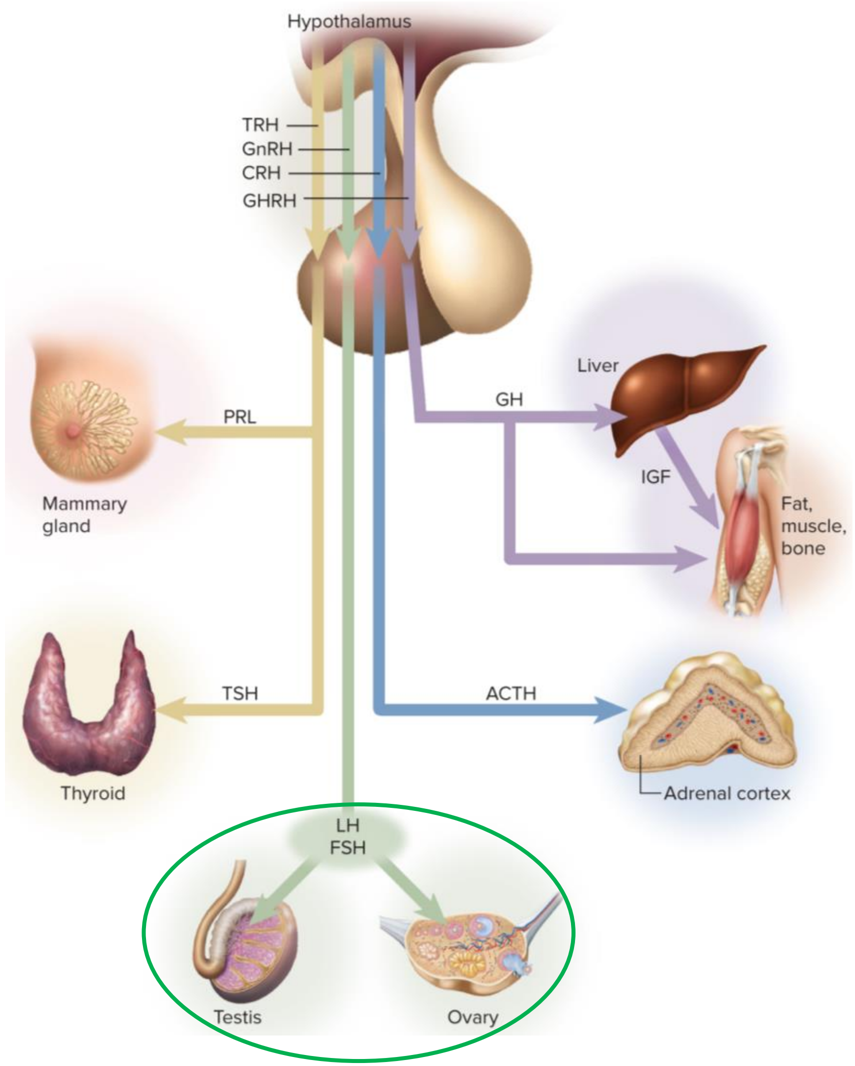
****follicle-stimulating hormone*** (***FSH***) stimulates the development of ovarian follicles and sperm production

The Hypothalamus and Pituitary Gland



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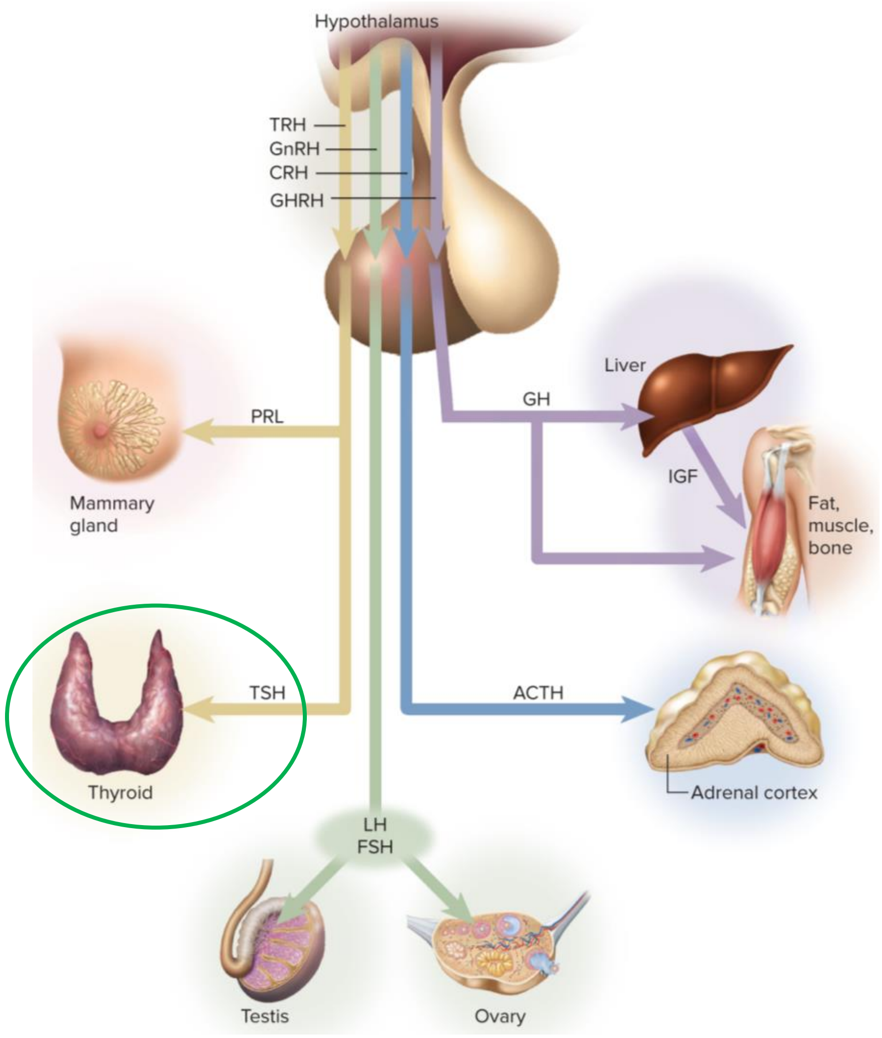
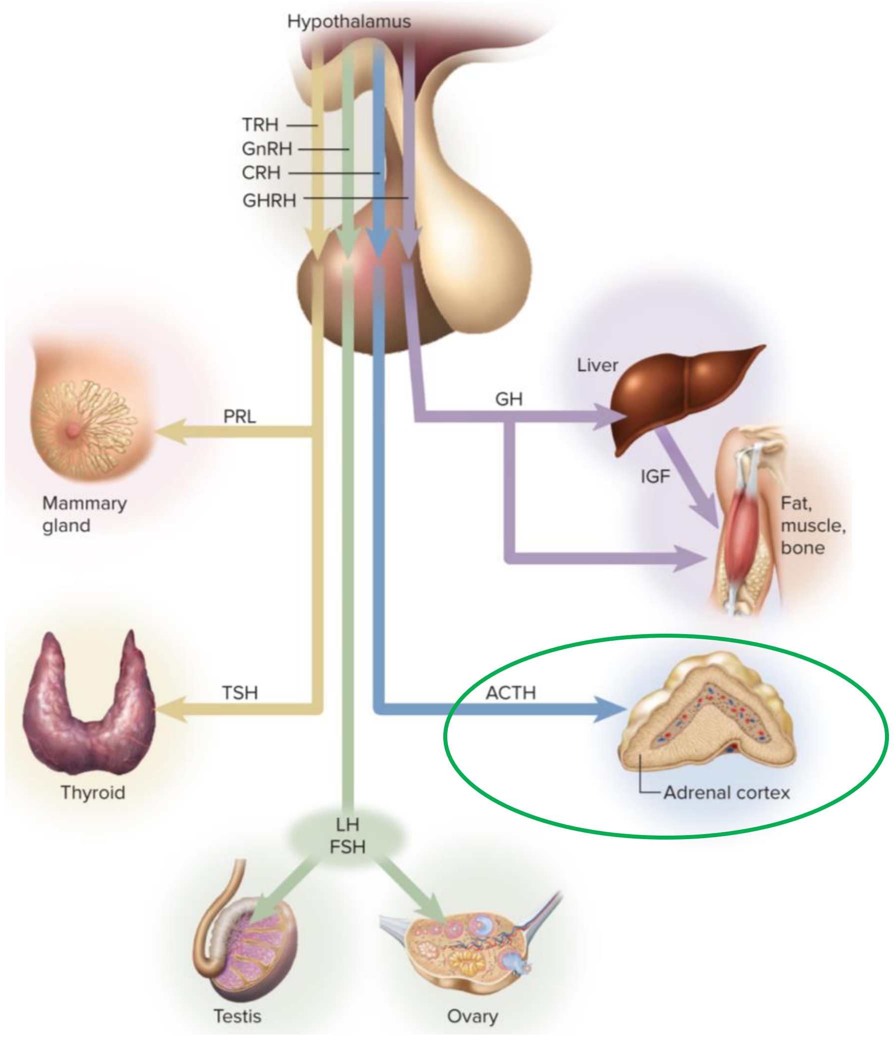
****luteinizing hormone*** (***LH***) stimulates ovulation in females and the secretion of testosterone in males

The Hypothalamus and Pituitary Gland



thyroid-stimulating hormone* (*TSH*), or ***thyro-tropin* stimulates the growth of the thyroid gland and secretion of thyroid hormone

The Hypothalamus and Pituitary Gland



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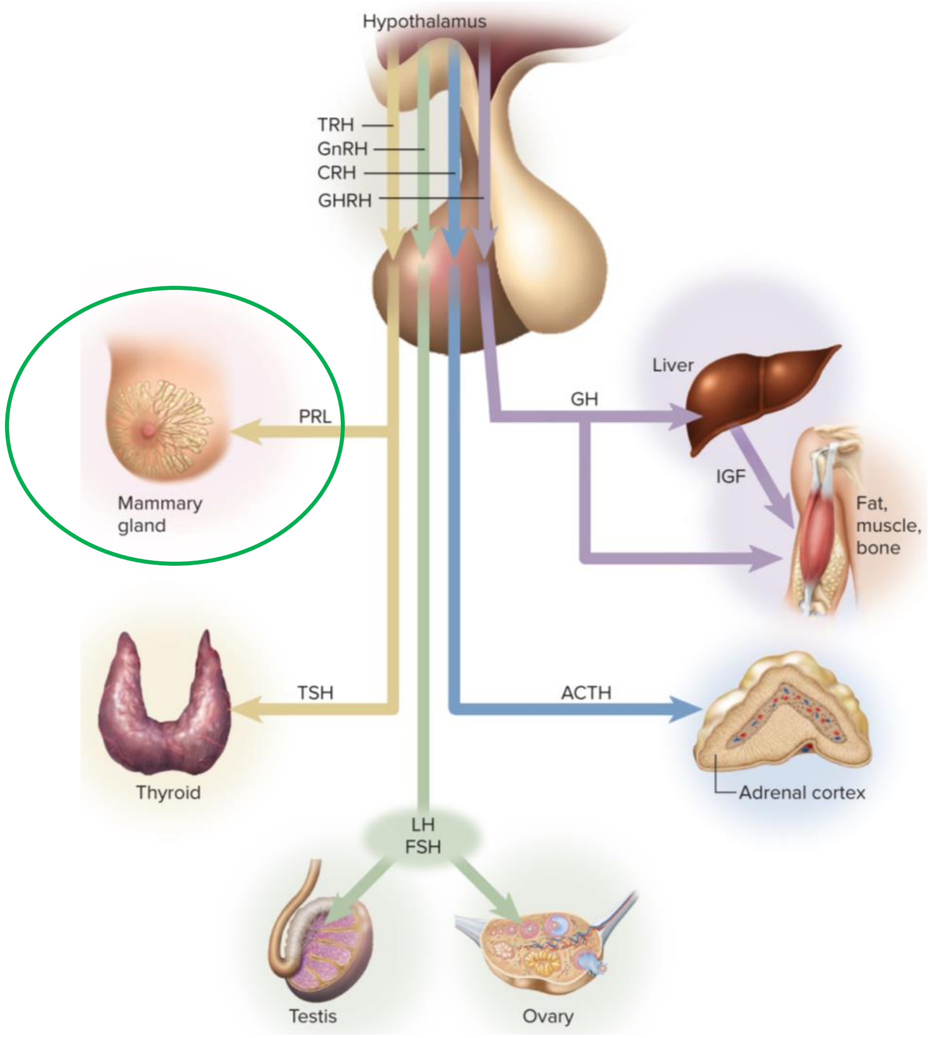

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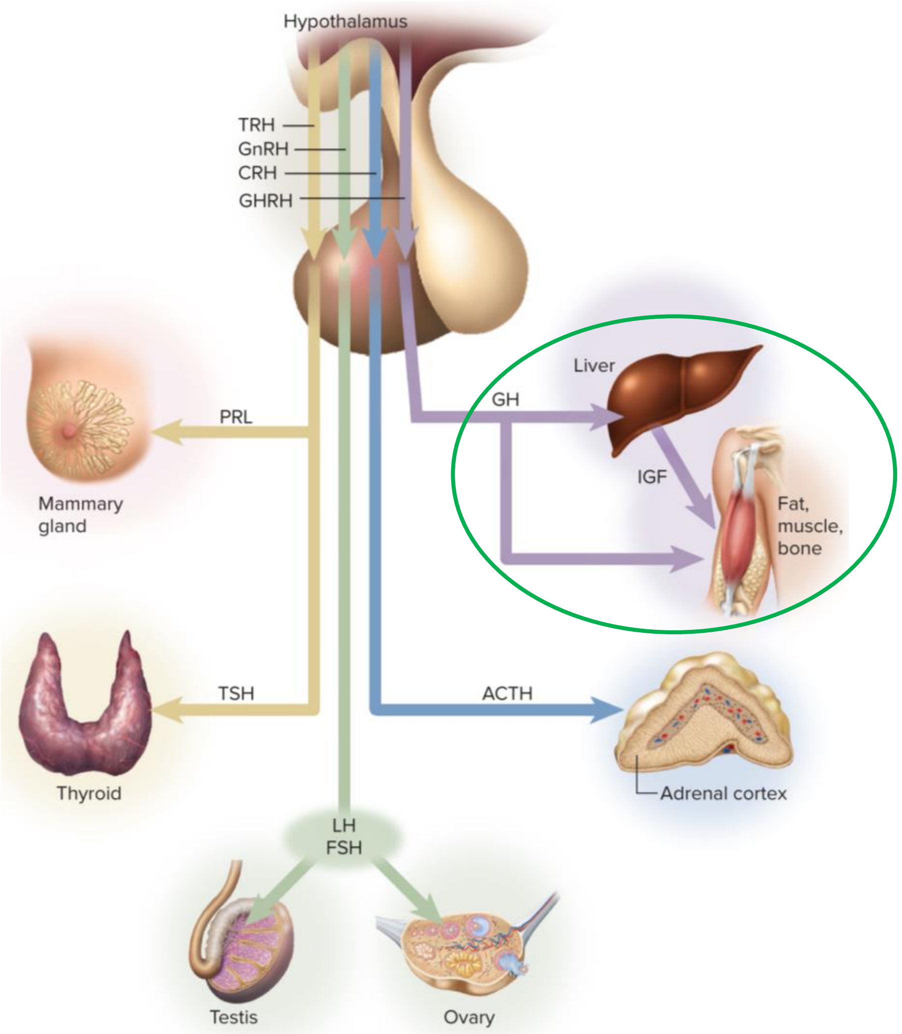
The Hypothalamus and Pituitary Gland



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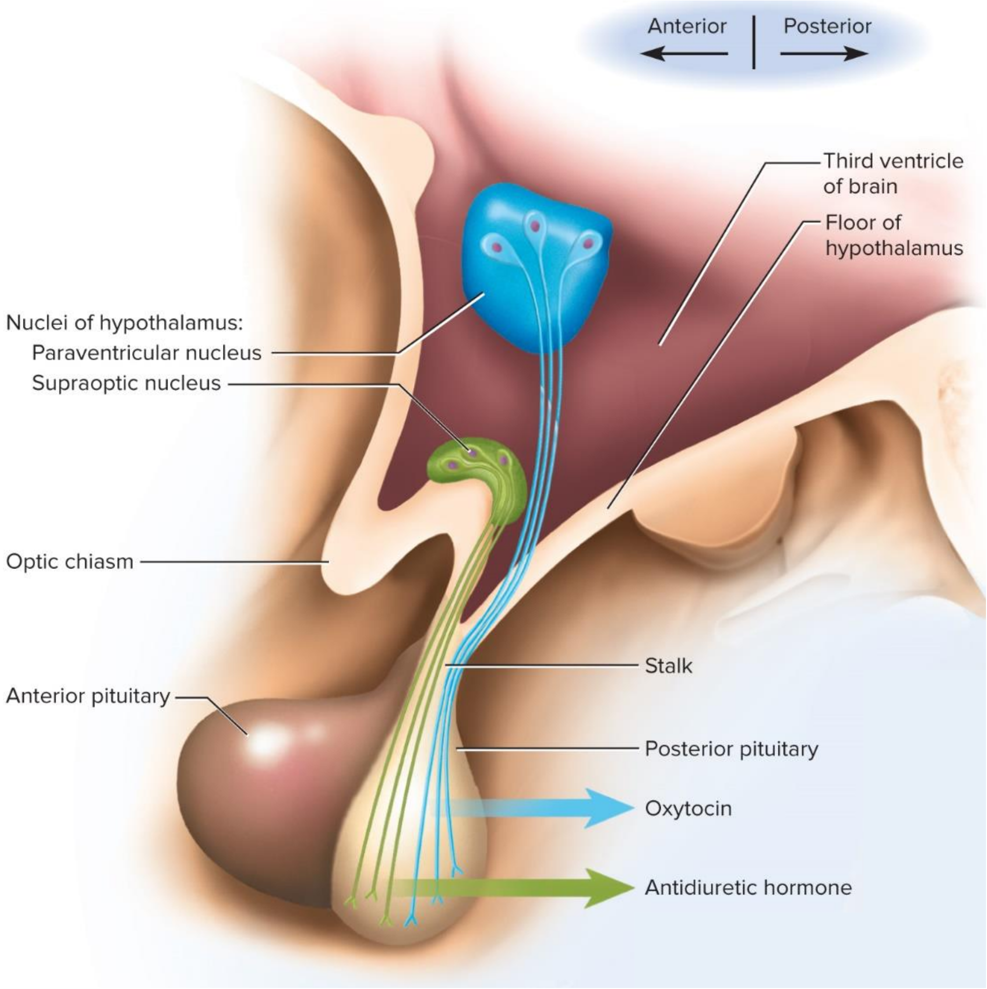
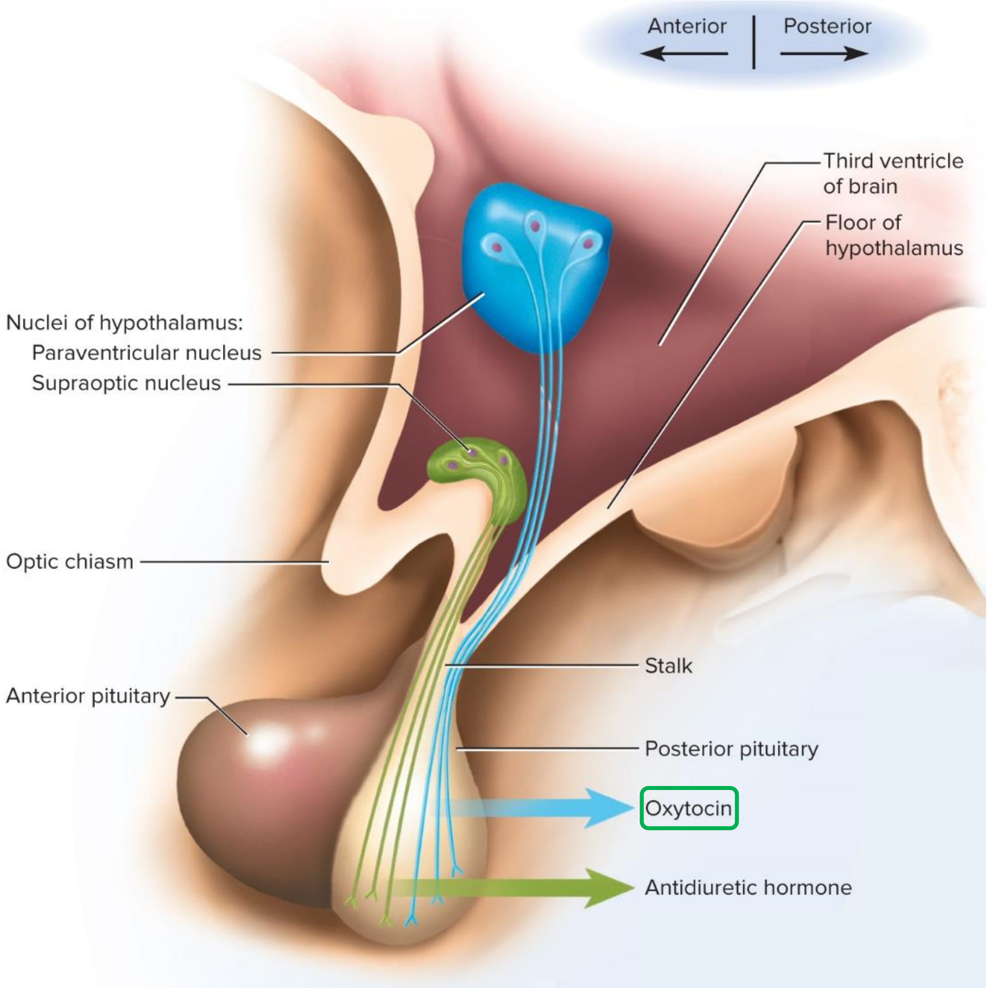
growth hormone** (GH***), or ***somatotropin*** stimulates mitosis and cellular differentiation

The Hypothalamus and Pituitary Gland



*the two posterior pituitary hormones are ***oxytocin*** (***OT***) and ***antidiuretic hormone*** (***ADH***)

The Hypothalamus and Pituitary Gland



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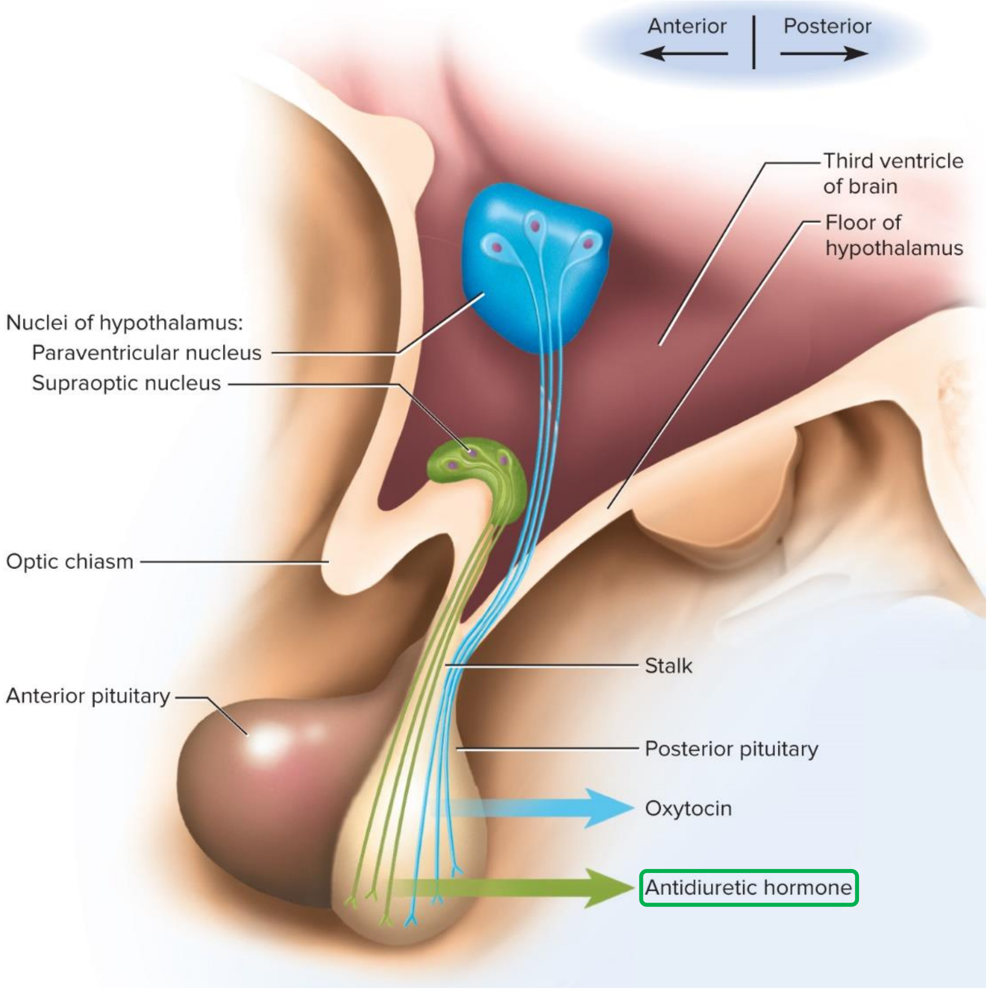
*oxytocin (OT) surges in both sexes during sexual arousal, stimulates labor contractions, and flow of milk during lactation, lowers stress, stimulates baby/mom bonding

The Hypothalamus and Pituitary Gland



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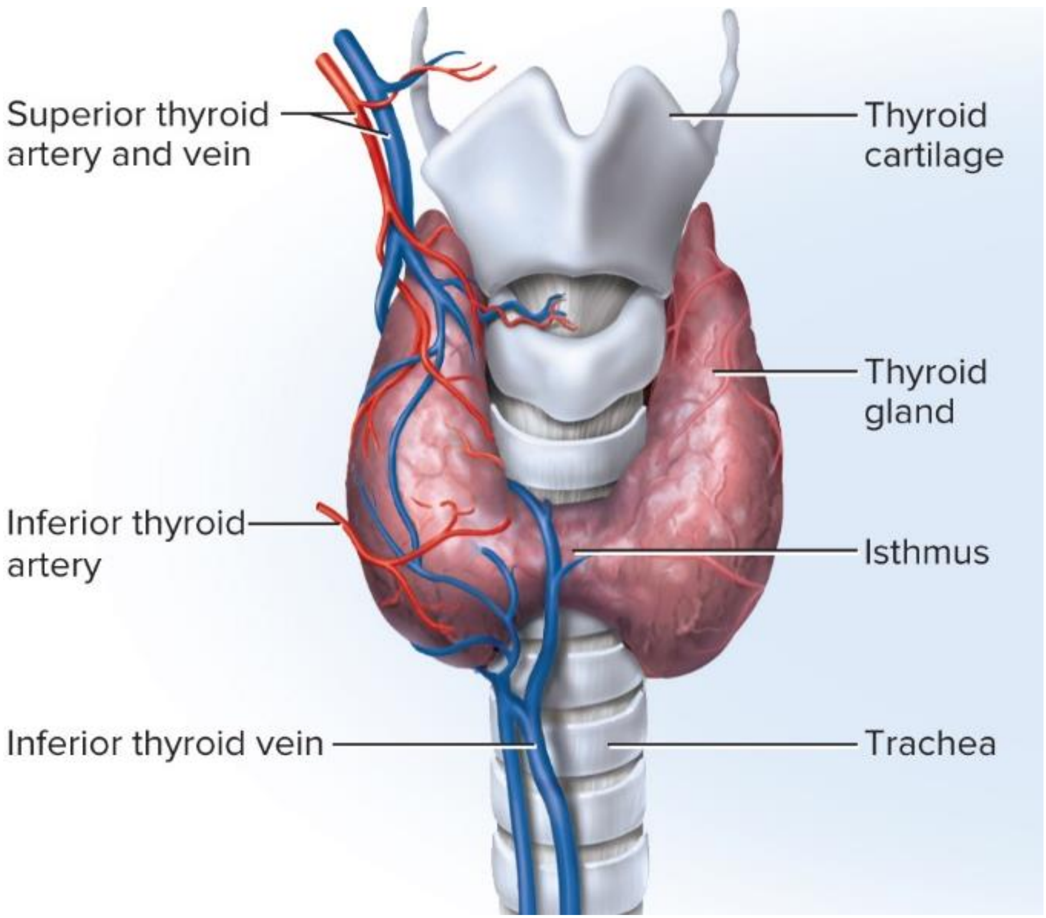
*oxytocin (OT) surges in both sexes during sexual arousal, stimulates labor contractions, and flow of milk during lactation, lowers stress, stimulates baby/mom bonding

*antidiuretic hormone (ADH) increases water retention by the kidneys, reduces urine volume, and helps prevent dehydration

The Thyroid Gland



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The Thyroid Gland

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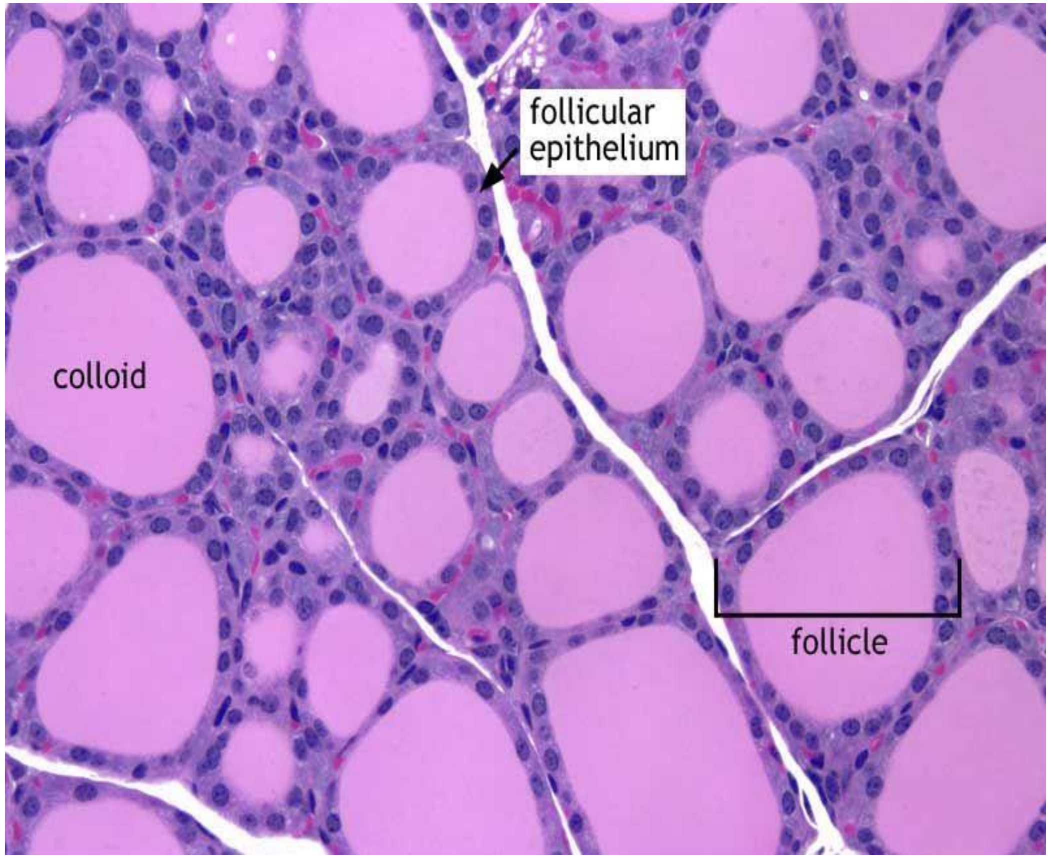
The Thyroid Gland



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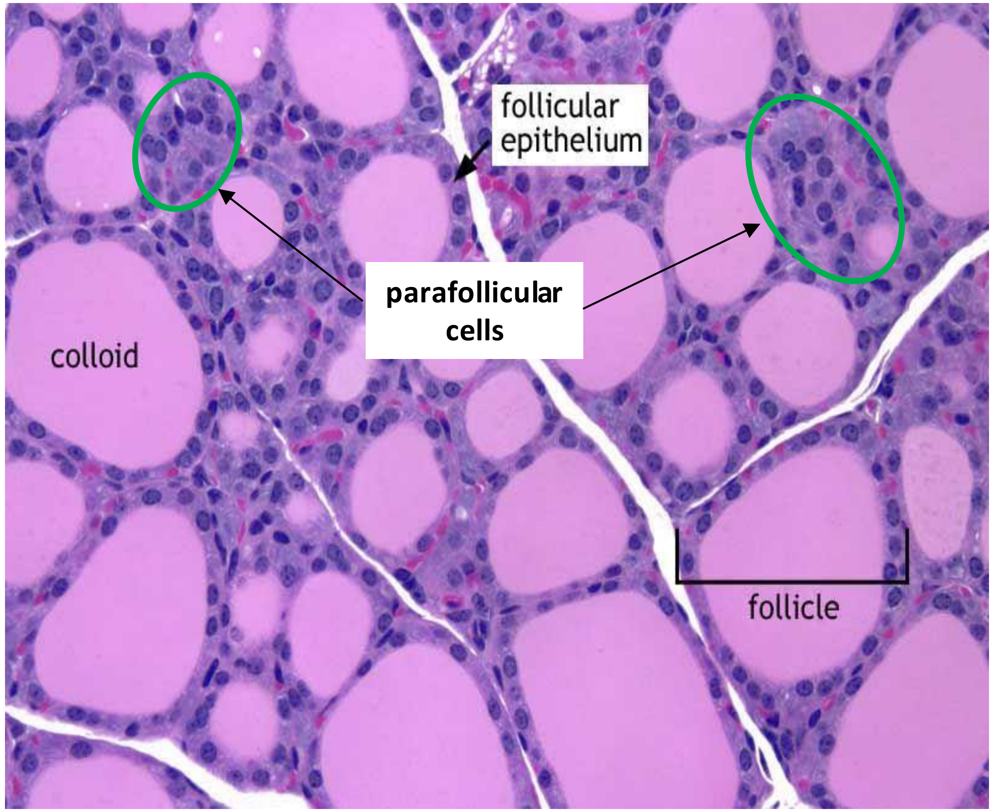
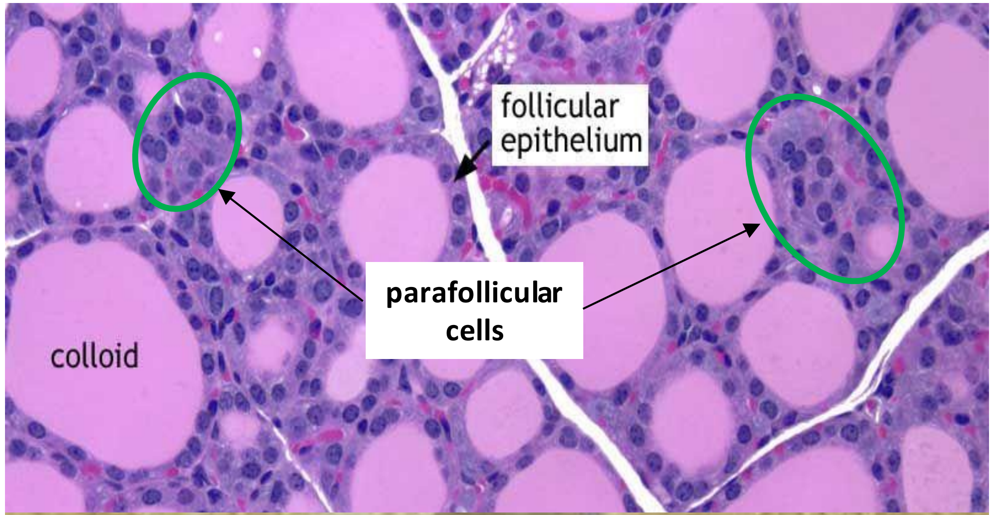
*thyroid follicles are filled with ***colloid*** and lined with ***follicular cells*** (secrete TH: $\boldsymbol{T_3}$ and $\boldsymbol{T_4}$). Thyroid hormone (TH) stimulates increased metabolic rate.

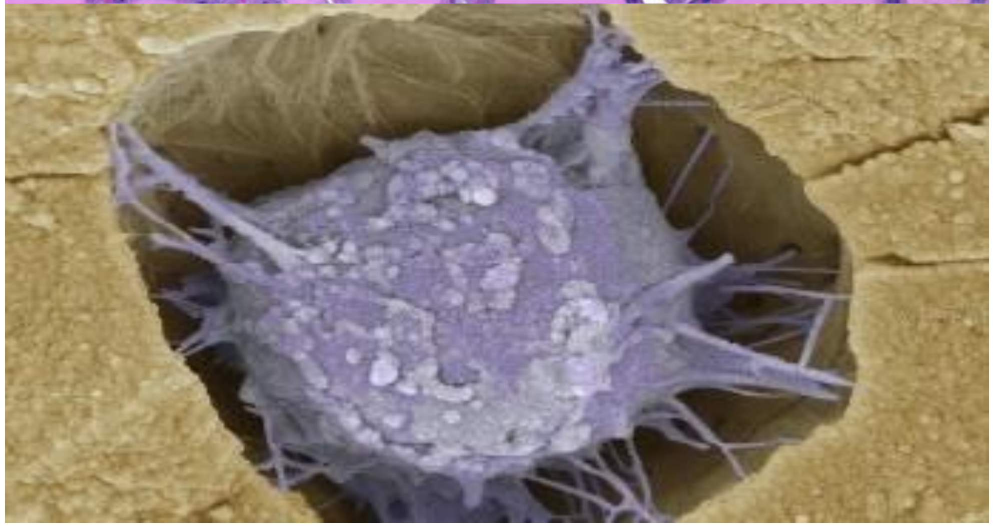
The Thyroid Gland



*thyroid gland also contains nests of ***parafol-licular cells*** or ***clear*** (***C***) cells at the periphery of the follicle

The Thyroid Gland

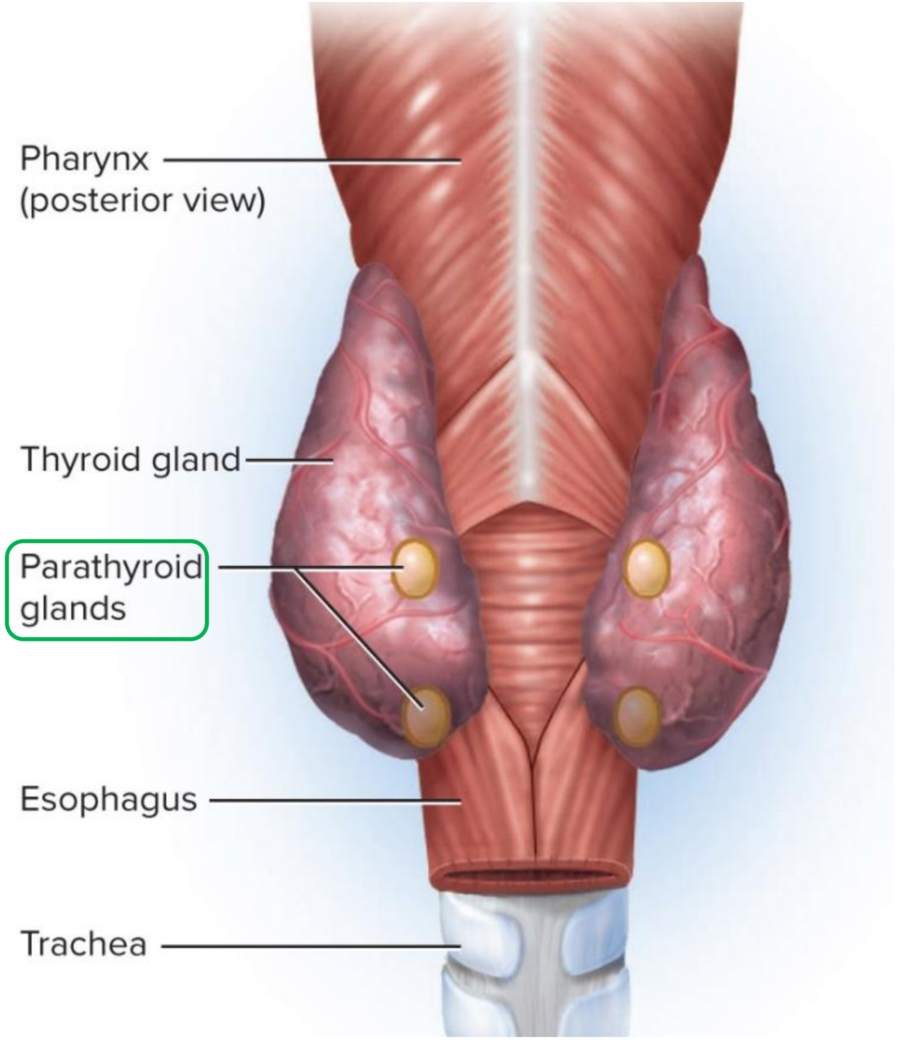




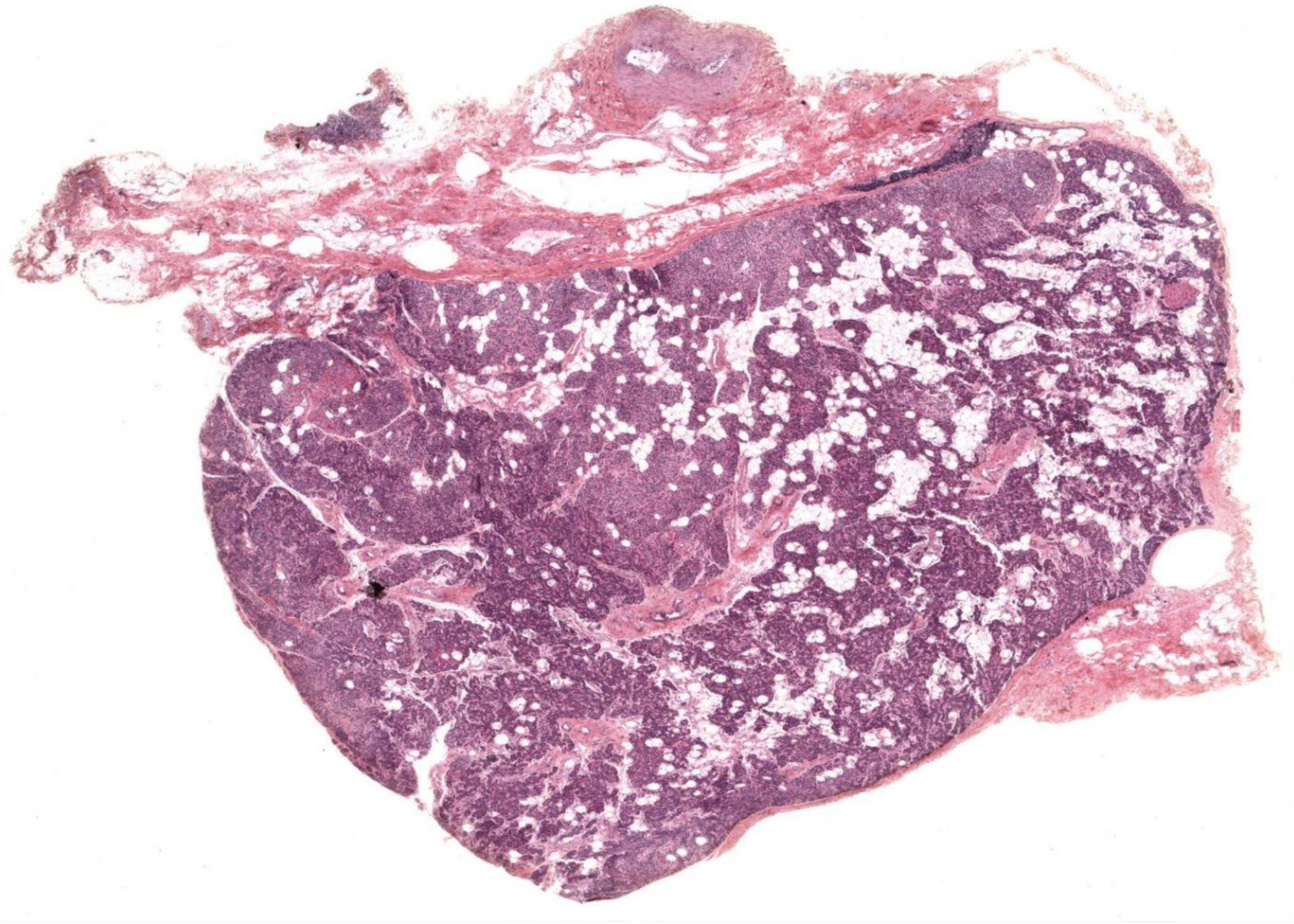
*thyroid gland also contains nests of ***parafol-
licular cells*** or ***clear*** (***C***) cells at the periphery of the follicle

*clear cells secrete ***calcitonin*** which decreases blood Ca^{++} level by stimulating osteoblast activity, promoting calcium deposition and bone formation in children

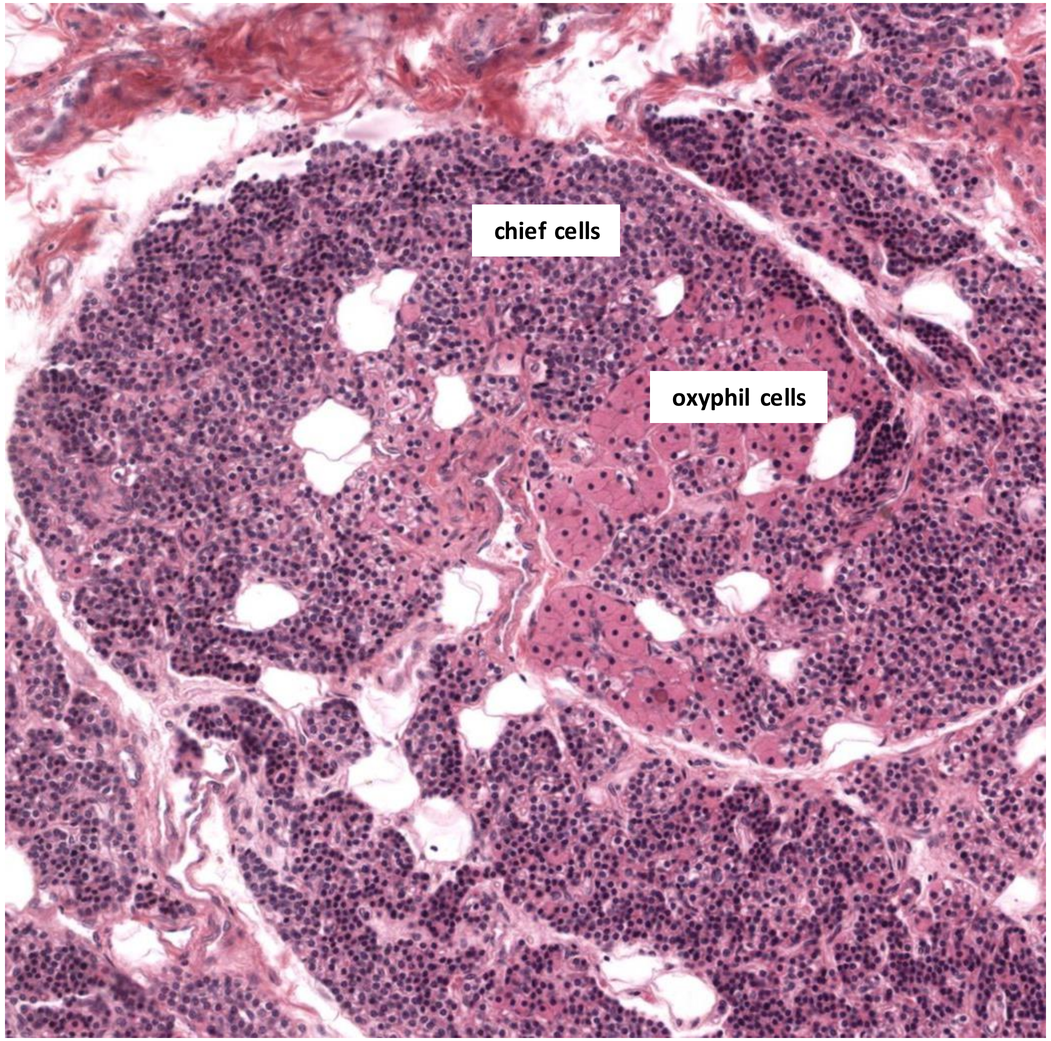
The Parathyroid Glands



**parathyroid glands* are ovoid glands, usually four in number, partially embedded in the posterior surface of the thyroid



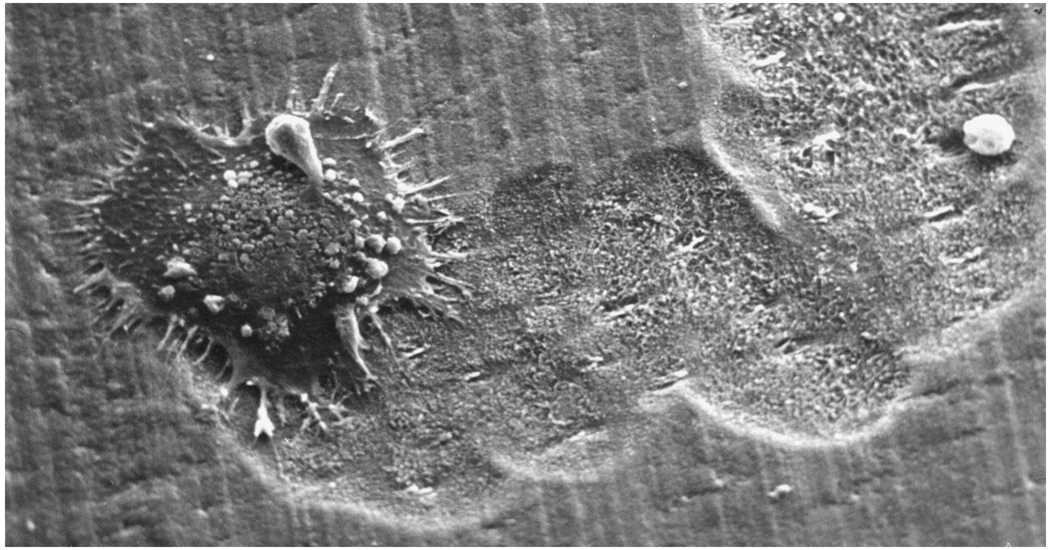
The Parathyroid Glands



*_**parathyroid glands**_ are ovoid glands, usually four in number, partially embedded in the posterior surface of the thyroid. Unlike the thyroid, parathyroid glands are not regulated but pituitary, but monitor blood directly.

*_**chief cells**_ secrete _**parathyroid hormone**_ (_**PTH**_), while _**oxyphil cells**_ are non secretory

The Parathyroid Glands

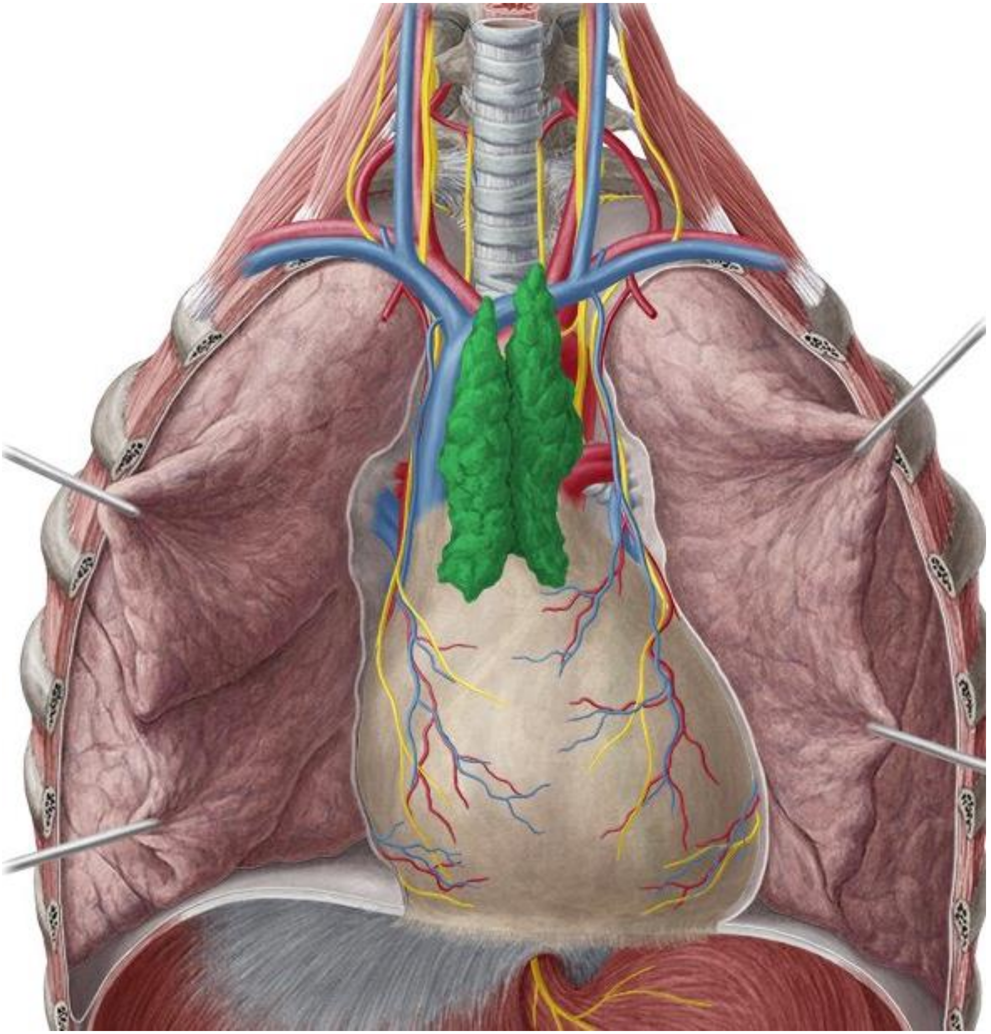


*__*parathyroid glands*__* are ovoid glands, usually four in number, partially embedded in the posterior surface of the thyroid

*__*chief cells*__* secrete ***parathyroid hormone*** (***PTH***), while ***clear cells*** are non secretory

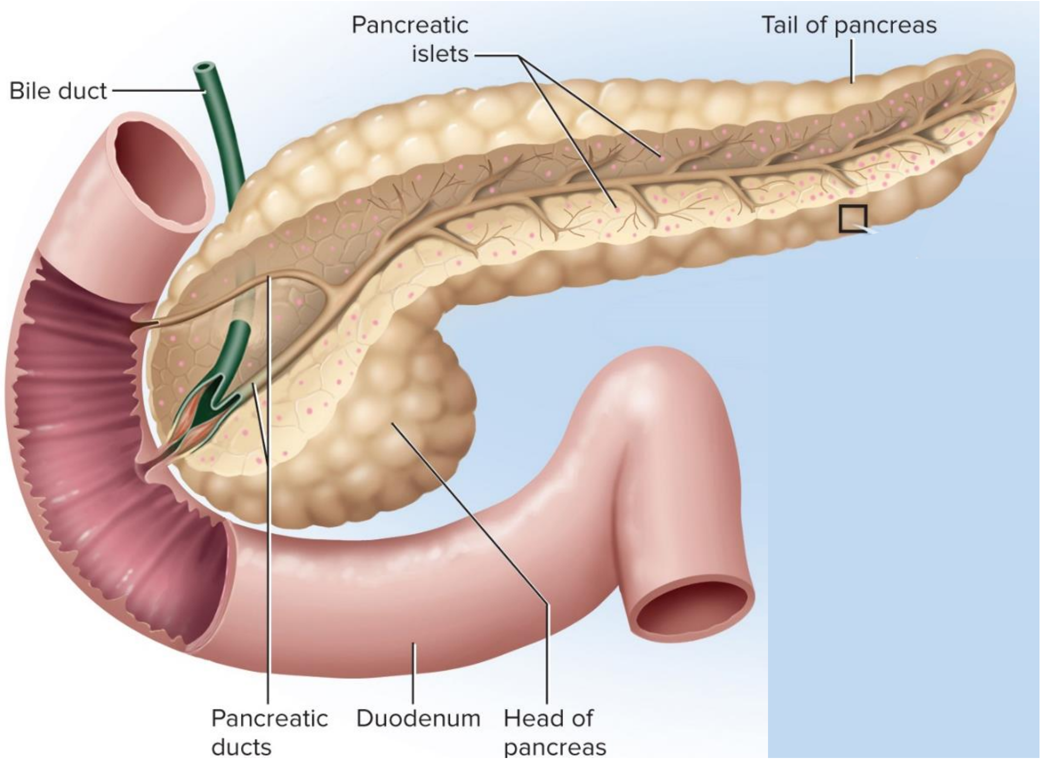
*PTH raises blood Ca^{2+} levels by stimulating Ca^{2+} reabsorption from the bones

The Thymus



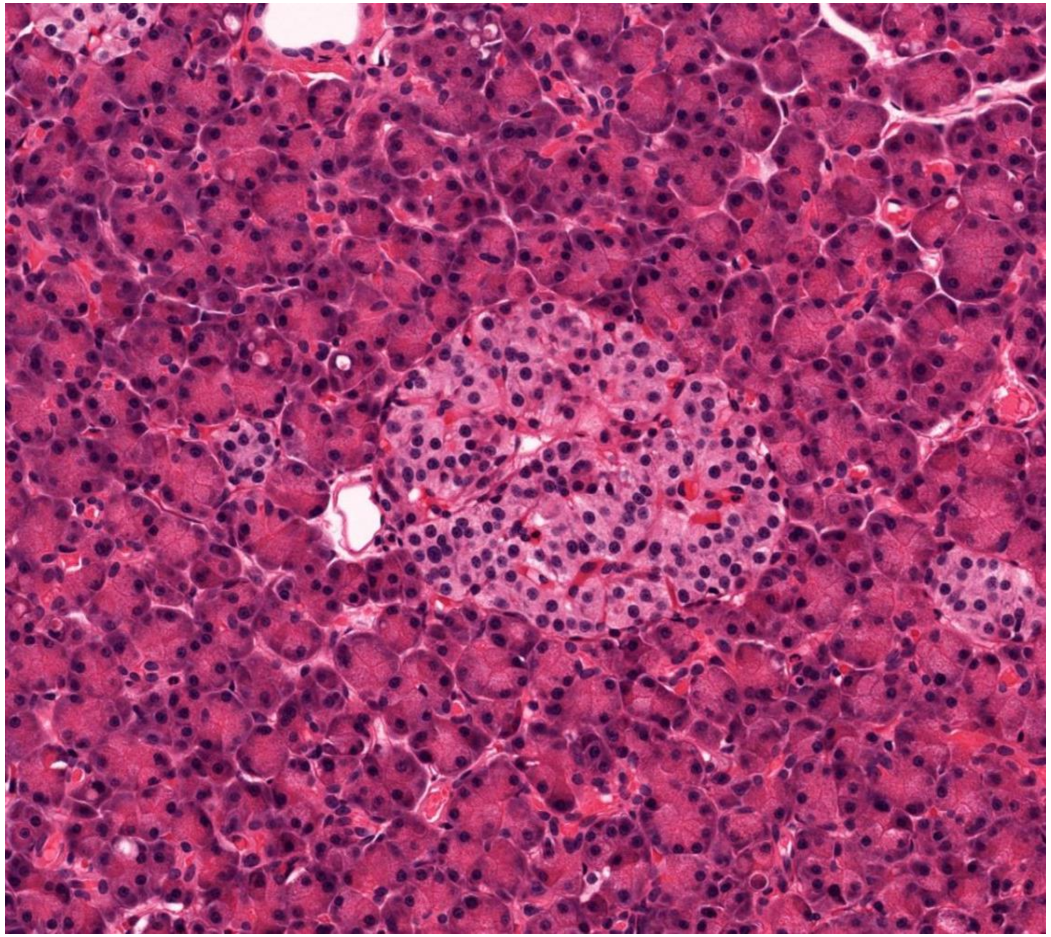
****thymus*** secretes ***thymosin,*** *which stimulates T-cell development and activity.*

The Pancreas



*pancreas is comprised of ***acinar cells*** (have exocrine function -- digestive enzymes) and ***pancreatic islet cells*** or ***islets of Langerhans***

The Pancreas

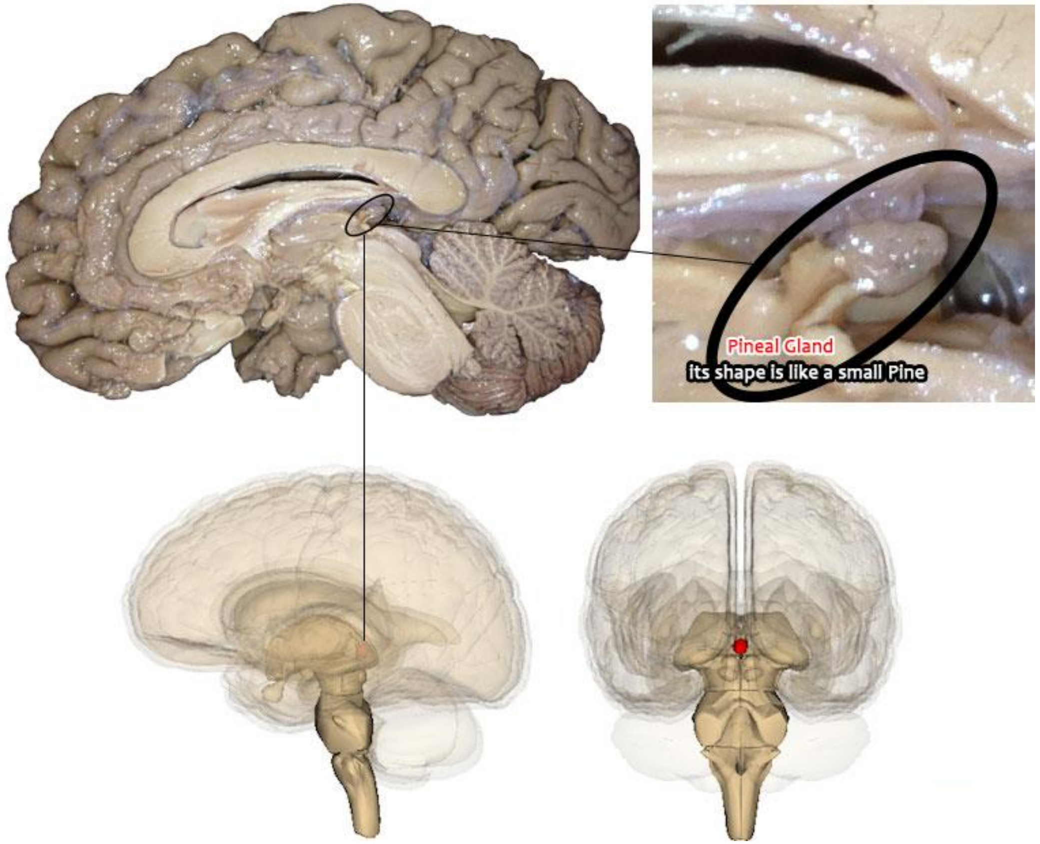


*pancreas is comprised of ***acinar cells*** (digestive enzymes) and ***pancreatic islet*** cells or ***islets of Langerhans***

Islet cells secrete insulin and glucagon

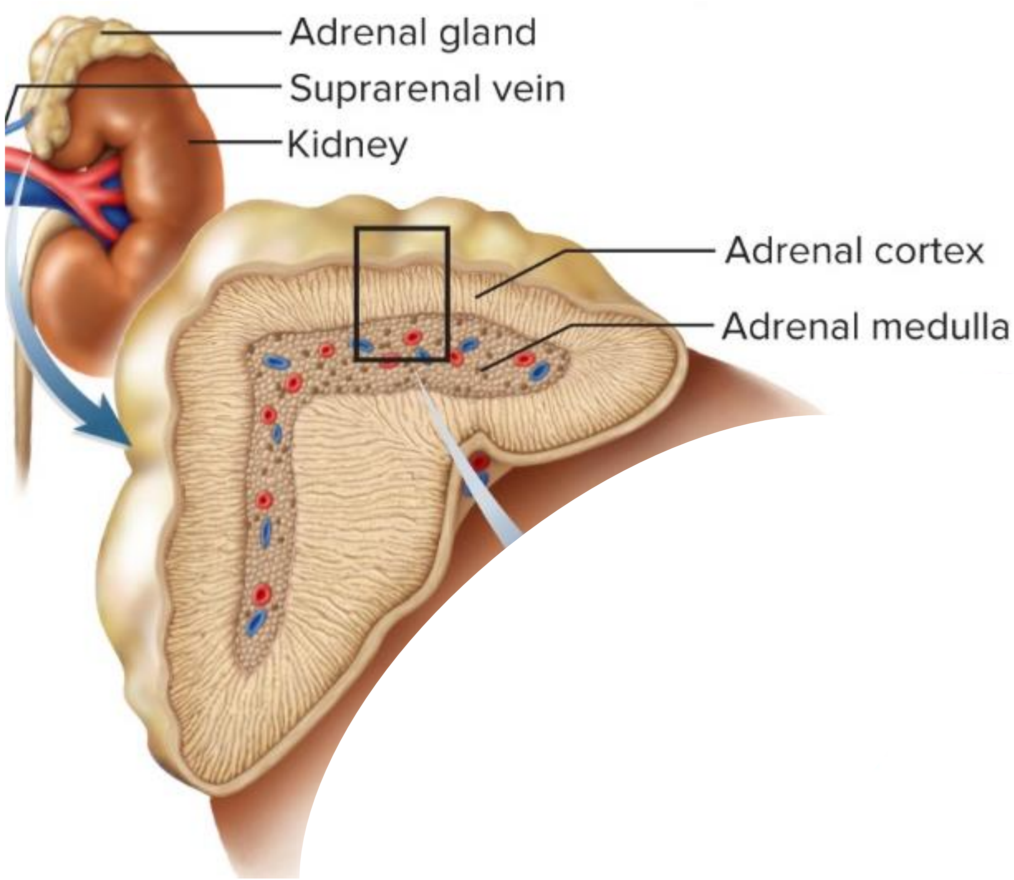
Insulin – secreted due to detected blood glucose level and stimulates glyconeogenesis in hepatocytes and lipogenesis in adipose tissue, while lipolysis, glycongenolysis and gluconeogenesis are inhibited.

The Pineal Gland



pineal gland is found in the epithalamus and secretes ***melatonin*** (sleep wake cycle)

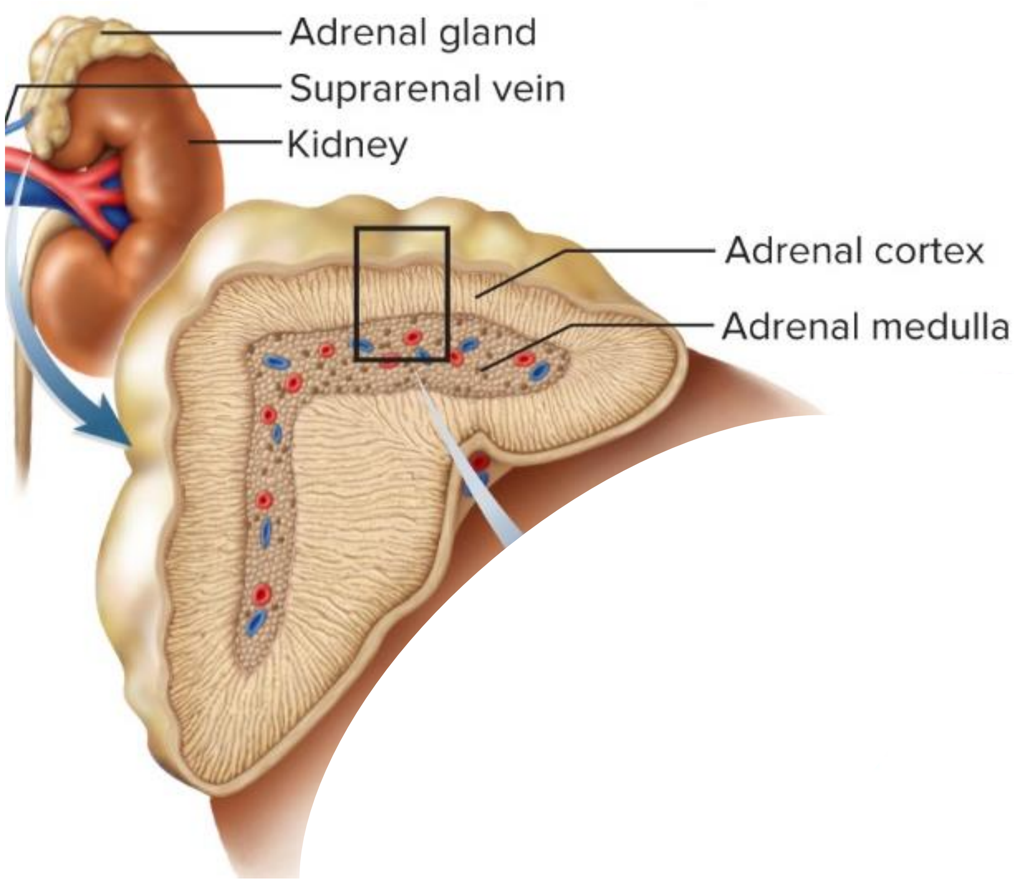
The Adrenal Glands



adrenal* (*suprarenal***) ***glands*** sit like a cap on the superior pole of each kidney

*comprised of an outer ***cortex*** (80-90%) and inner ***medulla*** (10-20%)

The Adrenal Glands



adrenal* (*suprarenal*) *glands* sit like a cap on the superior pole of each kidney

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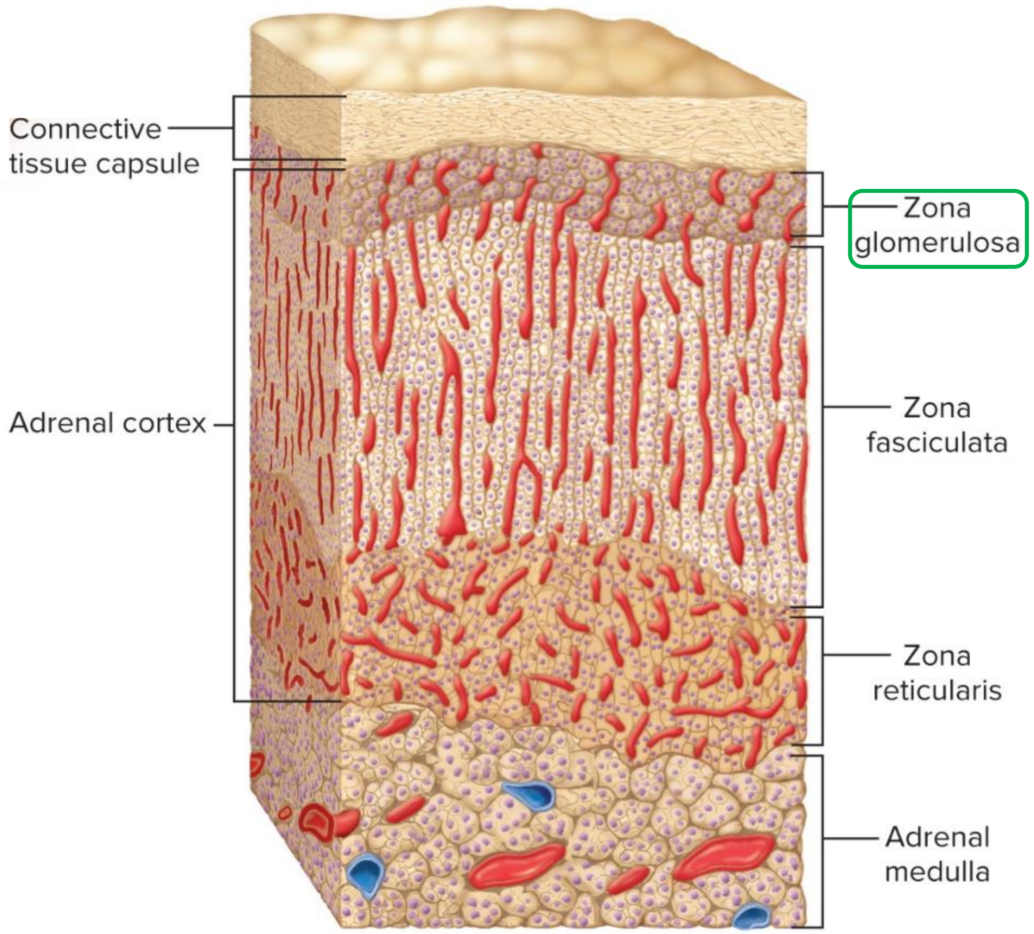
****adrenal medulla*** secretes ***epinephrine***

****adrenal cortex*** surrounds the medulla on all sides and produces more than 25 steroids

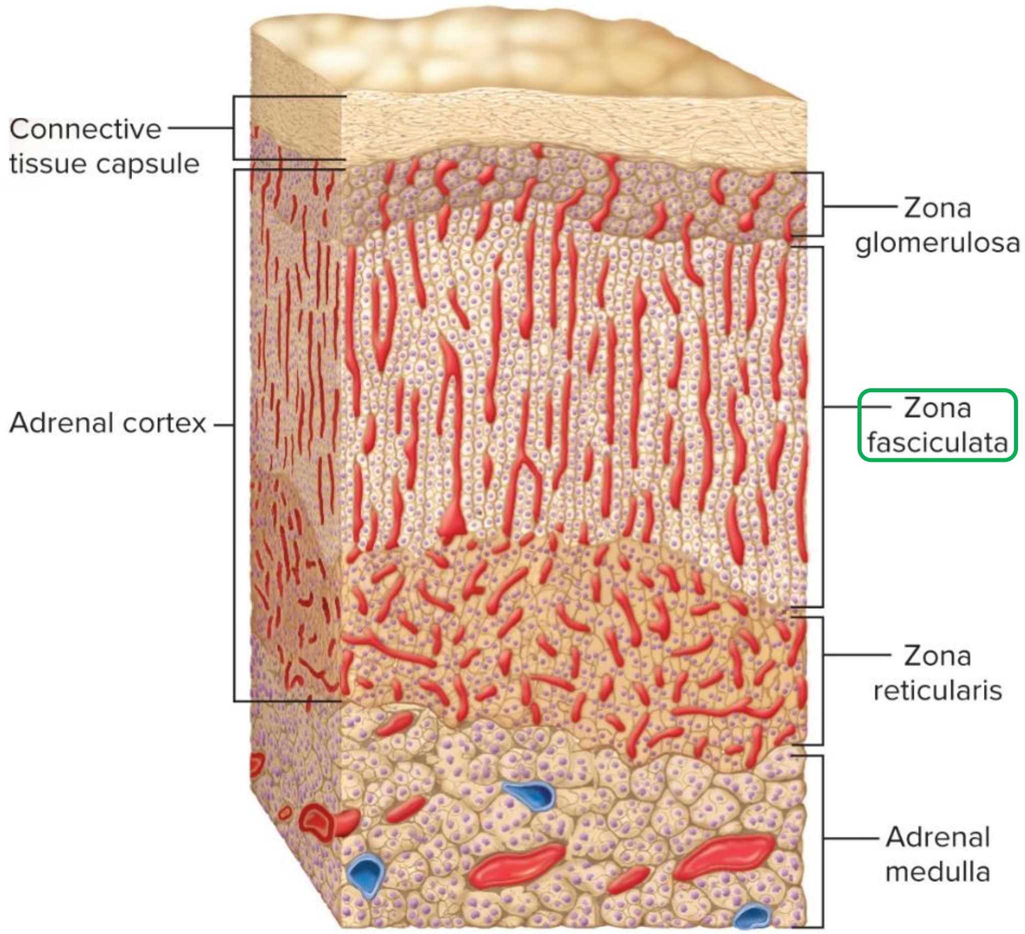
The Adrenal Glands



*adrenal cortex has three layers of tissue

****zona glomerulosa*** is the source of ***mineralocorticoids*** (***aldosterone*** – K+/Na+ balance)

The Adrenal Glands



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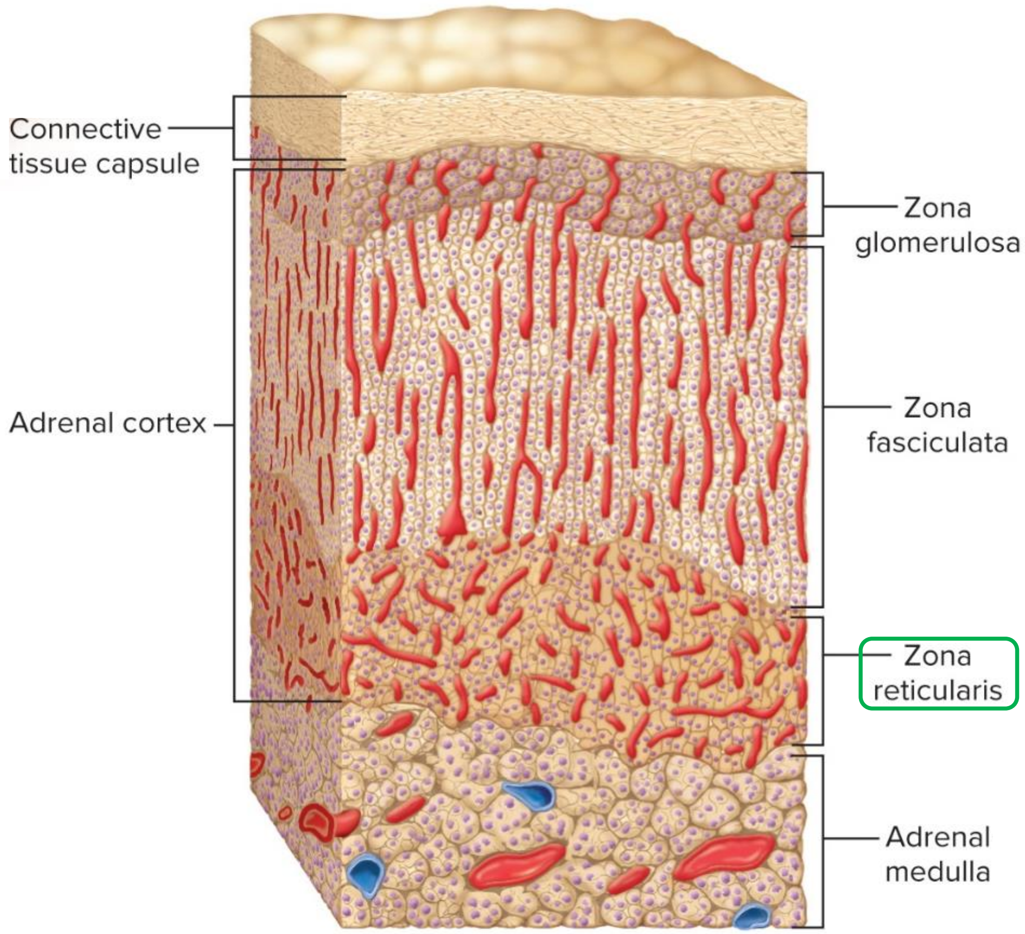
The Adrenal Glands



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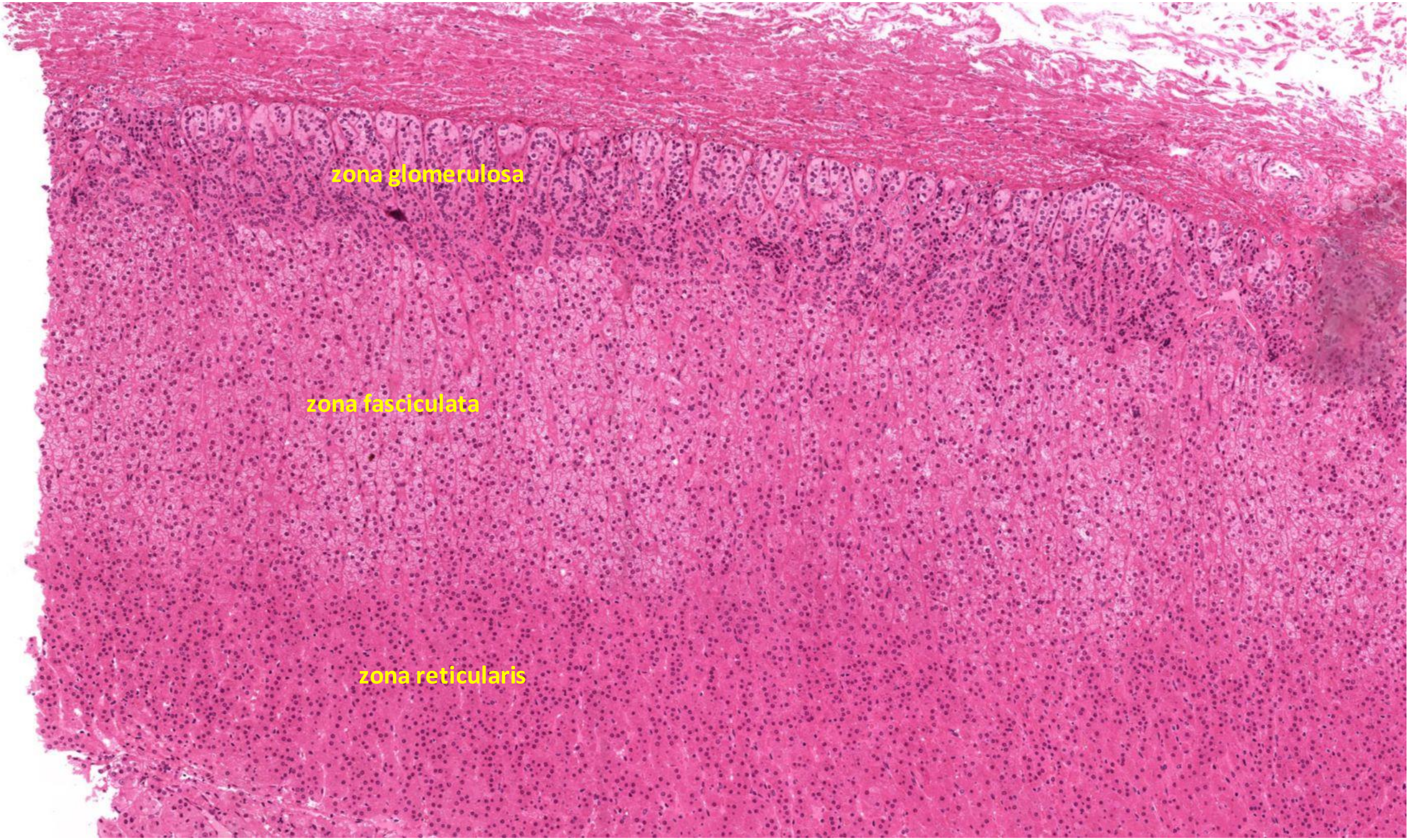
****zona fasciculata*** is the source of ***gluco-corticoids*** (***cortisol*** or ***hydrocortisone***)

****zona reticularis*** is the source of ***androgens***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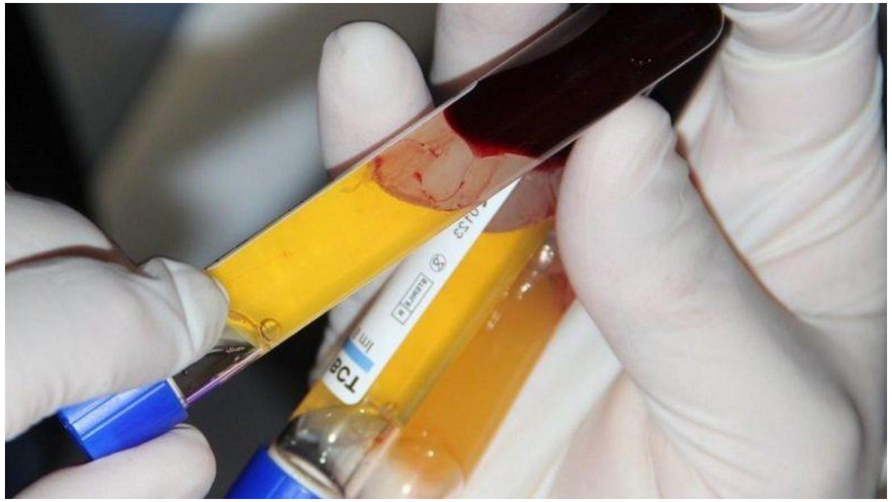
zona fasciculata
zona reticularis

Blood

*adults generally have 4-6 liters of blood

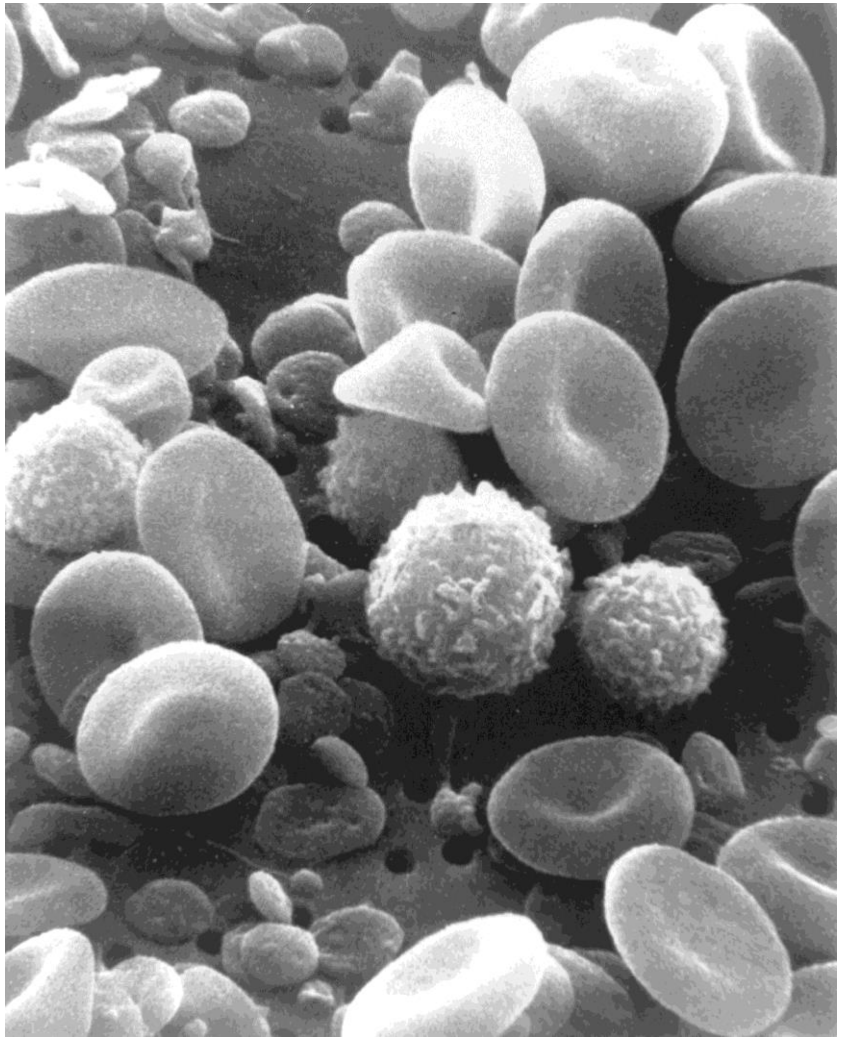
Blood



*adults generally have 4-6 liters of blood

*liquid connective tissue in which the matrix is blood ***plasma*** (clear, light yellow fluid)

Blood



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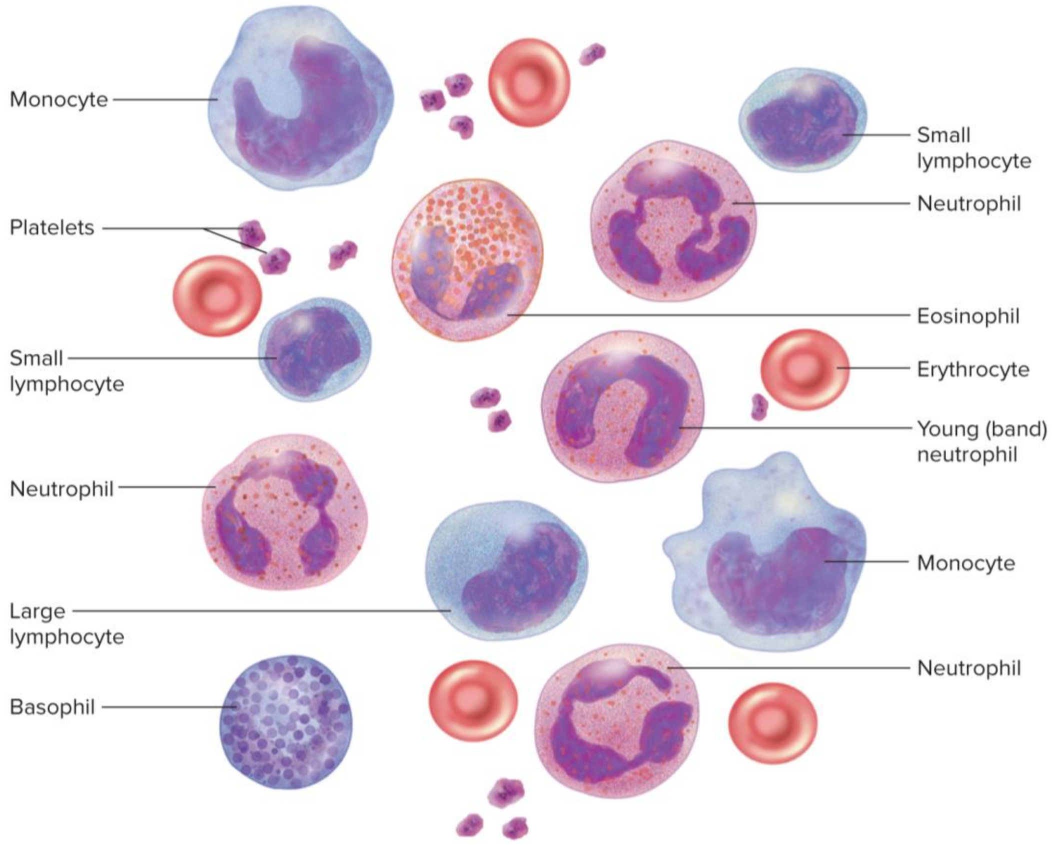
Blood



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*WBCs are divided into ***granulocytes*** (basophils, eosinophils, and neutrophils) and ***agranulocytes*** (monocytes and lymphocytes)

Blood

TABLE 18.1	General Properties of Blood
Characteristic	**Typical Values for Healthy Adults***
Mean fraction of body weight	8%
Volume in adult body	Female: 4–5 L; male: 5–6 L
Volume/body weight	80–85 mL/kg
Mean temperature	38°C (100.4°F)
pH	7.35–7.45
Viscosity (relative to water)	Whole blood: 4.5–5.5; plasma: 2.0
Osmolarity	280–296 mOsm/L
Mean salinity (mainly NaCl)	0.9%
Hematocrit (packed cell volume)	Female: 37% to 48% Male: 45% to 52%
Hemoglobin	Female: 12–16 g/dL Male: 13–18 g/dL
Mean RBC count	Female: 4.2–5.4 million/µL Male: 4.6–6.2 million/µL
Platelet count	130,000–360,000/µL
Total WBC count	5,000–10,000/µL

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*liquid connective tissue in which the matrix is blood ***plasma*** (clear, light yellow fluid)

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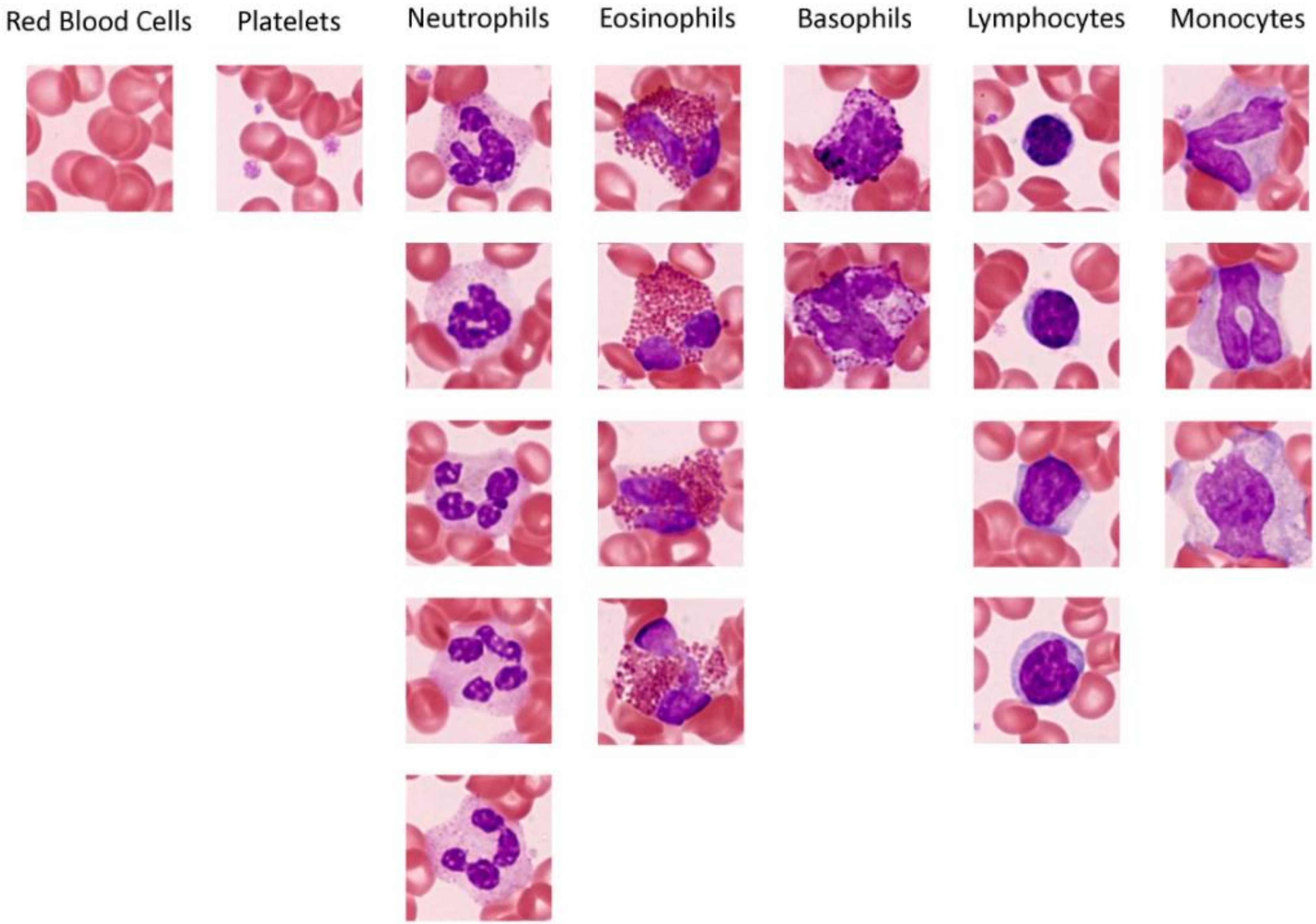
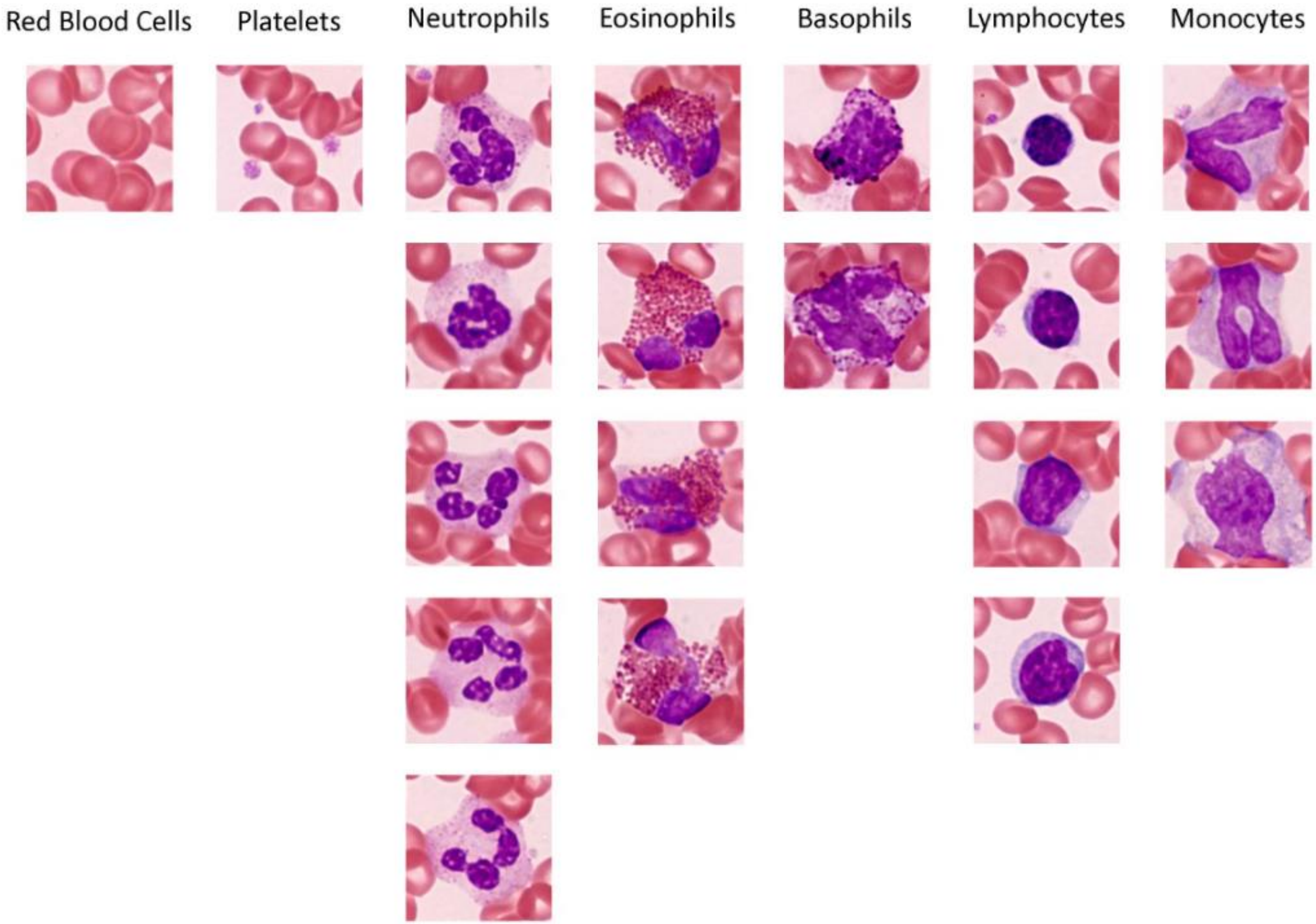

Red Blood Cells
Platelets
Neutrophils
Eosinophils
Basophils
Lymphocytes
Monocytes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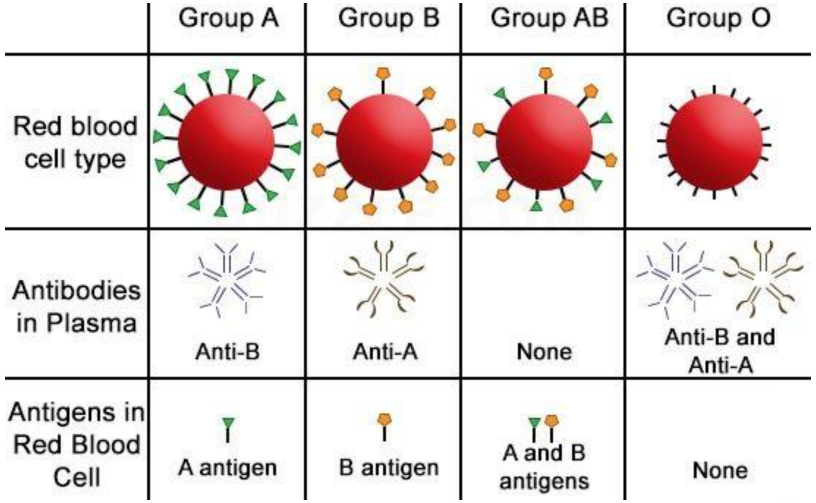
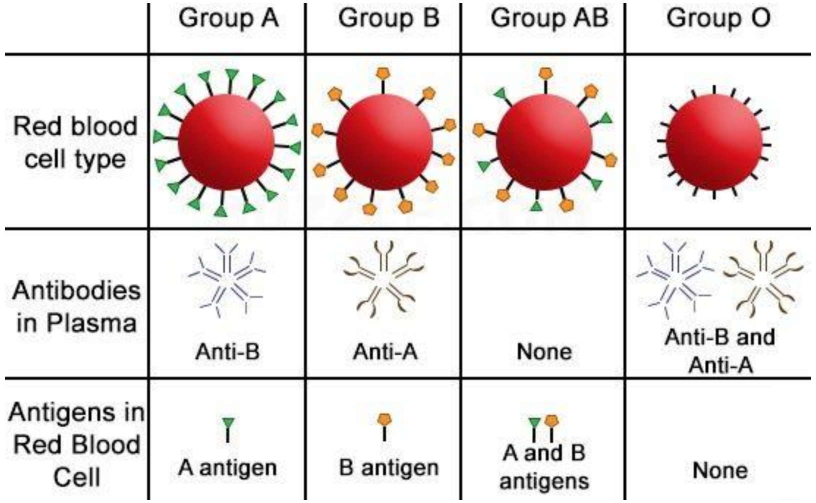
Blood Types



*type O blood is the most common and AB is the rarest in the U.S.

Blood Types



*type O blood is the most common and AB is the rarest in the U.S.

*in giving transfusions, it is imperative that the donor's RBCs not agglutinate as they enter the recipient's bloodstream

*a mismatch causes a ***transfusion reaction***
- agglutinated RBCs block small blood vessels, hemolyze, and release their Hb

*type AB is called the ***universal recipient*** and type O the ***universal donor***